Data science lifecycle & Exploratory data analysis using visualization

ds4owd - data science for openwashdata

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ds4owd-002.github.io/website/

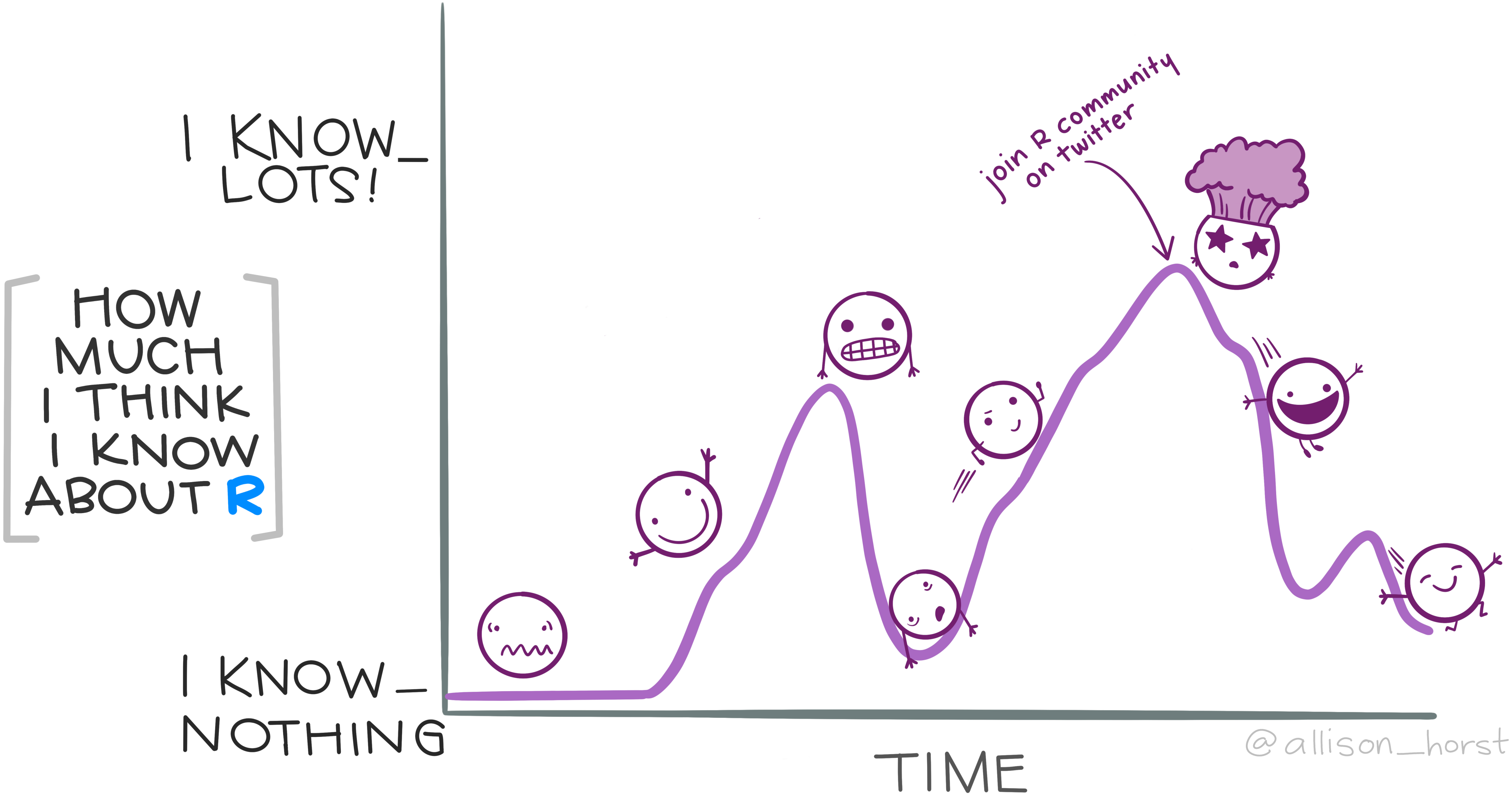
Q: How do I successfully complete the course?

You successfully complete the course and youwill receive a certificate of completion if you:

1. Complete the quiz of each week by its due date

2- Hand in a complete capstone project report that uses a dataset of your choice by 11 December 2025 (instructions will follow)

While homework assignments are not required for completion, we highly recommend working on them and submitting it for


I KNOW LOTS!
HOW MUCH I THINK I KNOW ABOUT R
I KNOW NOTHING
TIME
join R community on twitter
@allison_horst

Solving coding problems online

Tipps for search engines (e.g. duckduckgo.com)

- Use actionable verbs that describe what you want to do
- Be specific
- Add R to the search query
- Add the name of the R package name to the search query
- Scroll through the top 5 results (don’t just pick the first)

Example: “How to remove a legend from a plot in R ggplot2”

Stack Overflow

What is it?

- The biggest support network for (coding) problems
- Can be intimidating at first
- Up-vote system

Workflow

- First, briefly read the question that was posted
- Then, read the answer marked as “correct”
- Then, read one or two more answers with high votes

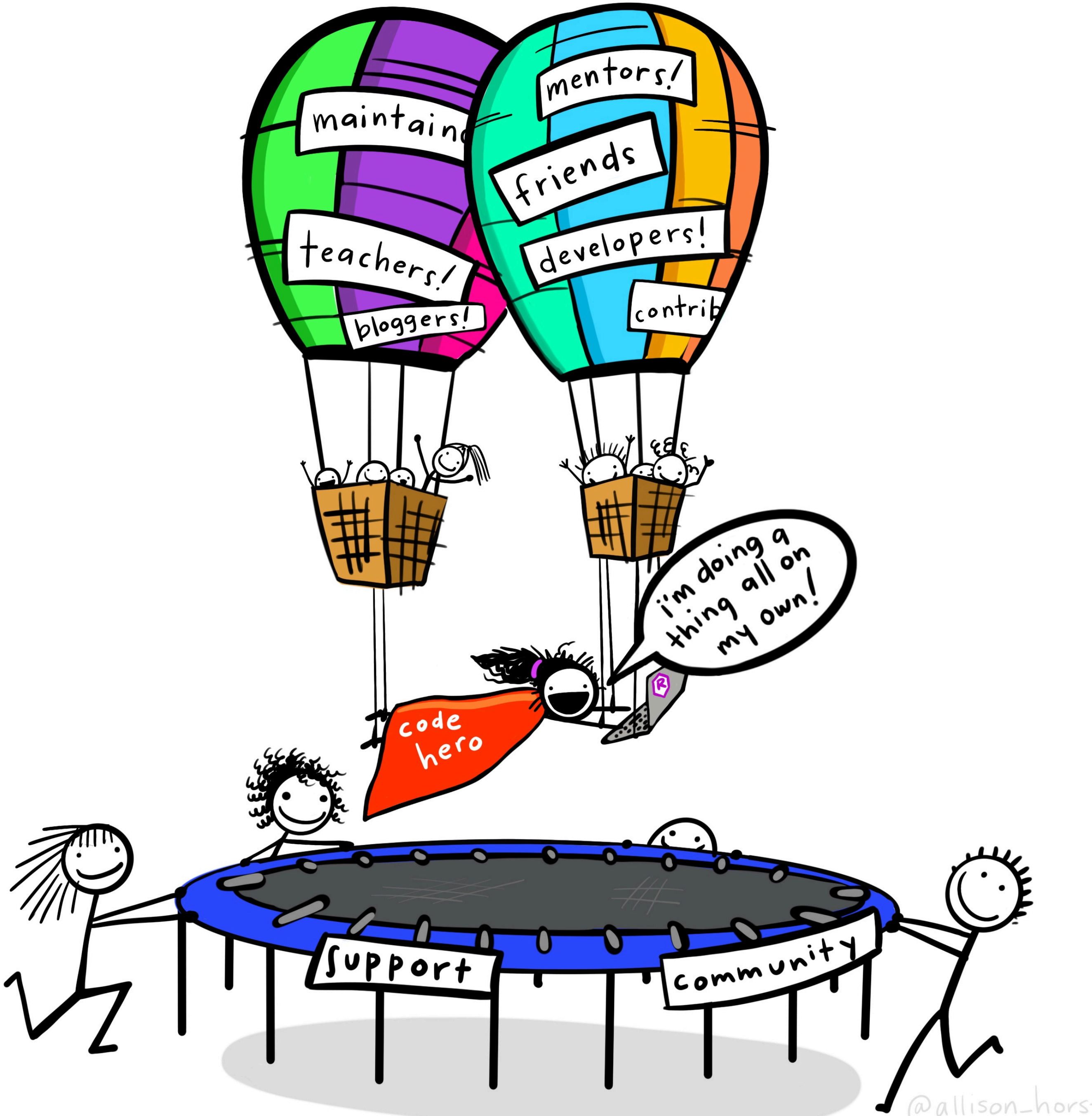
Tipps for AI tools (e.g. duck.ai)

- Use actionable verbs that describe what you want to do
- Be specific
- Add R to the search query
- Add the name of the R package name to the search query

Example: “How to remove a legend from a plot in R ggplot2”

Other sources for help

- Posit Community Forum: https://community.rstudio.com/
- Documentation websites: https://ggplot2.tidyverse.org/
- Mastodon / Bluesky tag: #rstats



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Exercises & Assignments

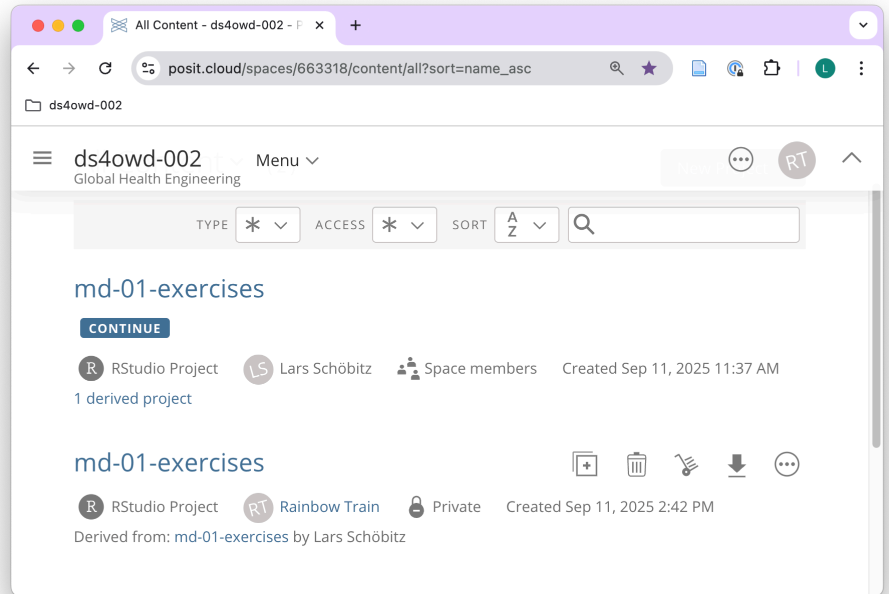
Note

For every week, there will be two “projects”:

- md-0X-exercises (this is for lectures and we only work with it during lectures)
- md-0X-assignments-USERNAME (this is your homework and every module will have your username at the end of it. You will clone this to Posit Cloud to work with it)

Lecture Exercises

- Click Start the first time
- Click Continue the next time
- It will look like two projects, but it is just one
- It doesn't matter on which of the two you click



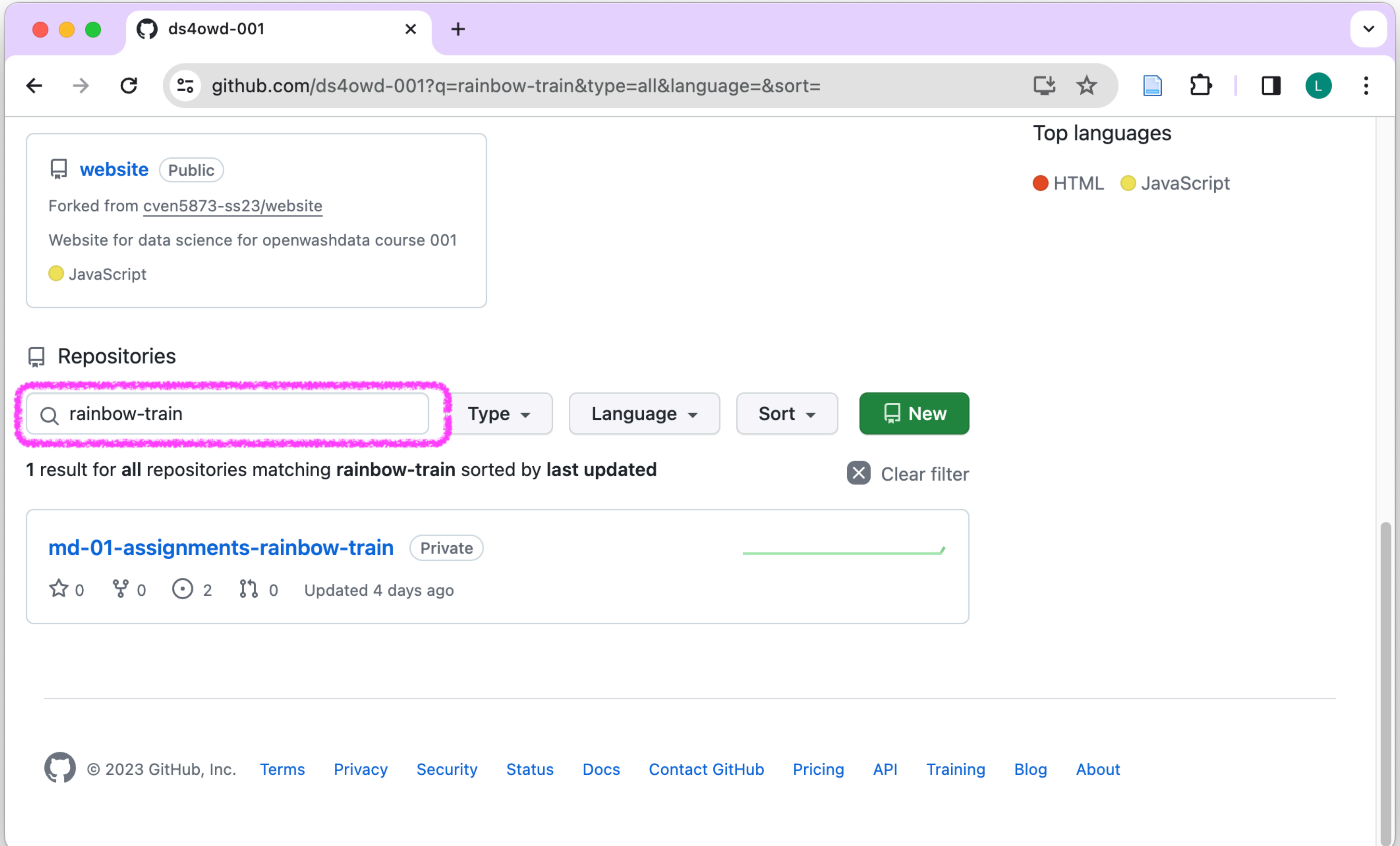
openwashdata.

Homework Assignments

on GitHub Organisation

Assignment 1 of Module 2 is a Bookmark Folder assignment!

https://github.com/ds4owd-002



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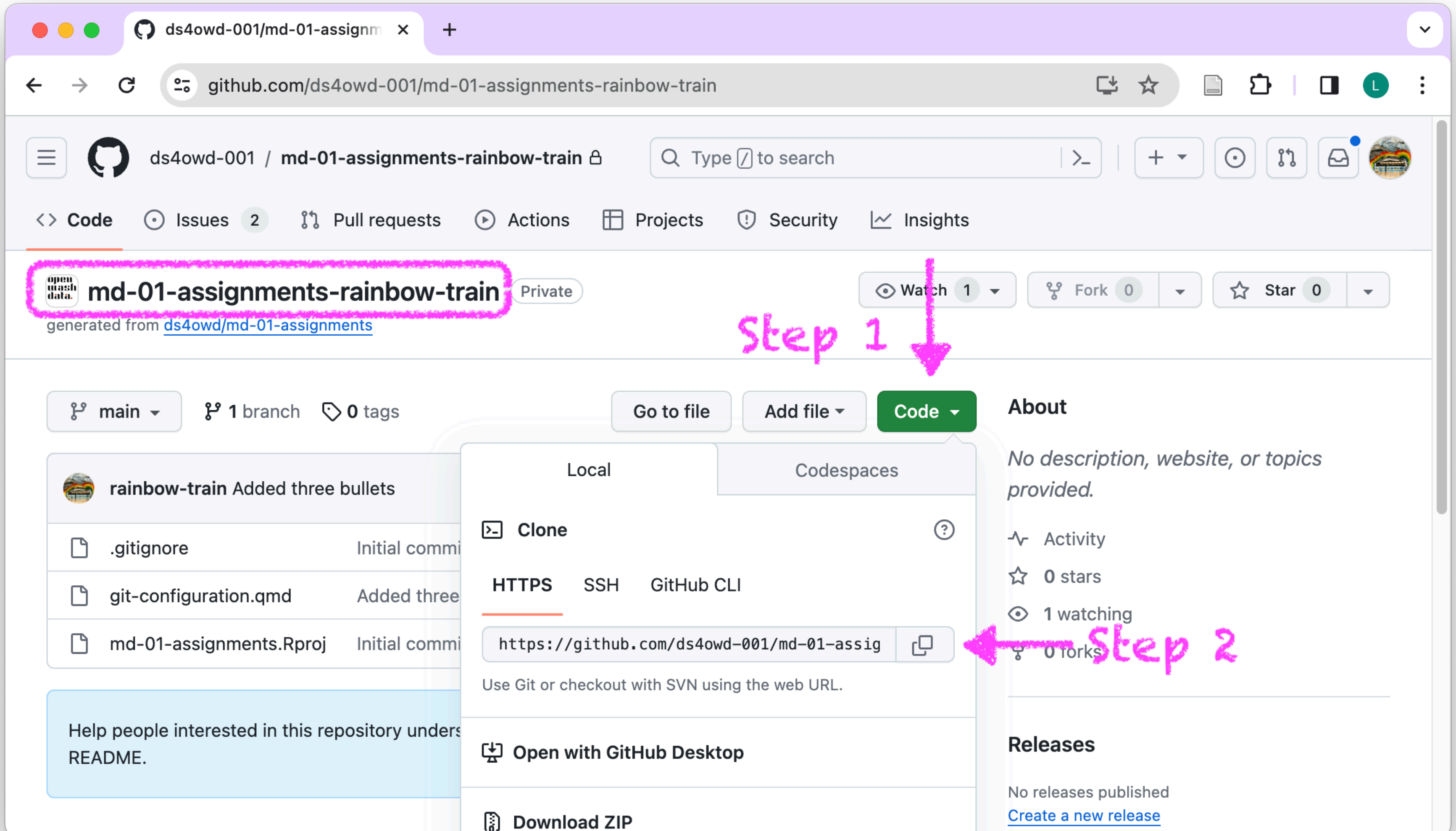
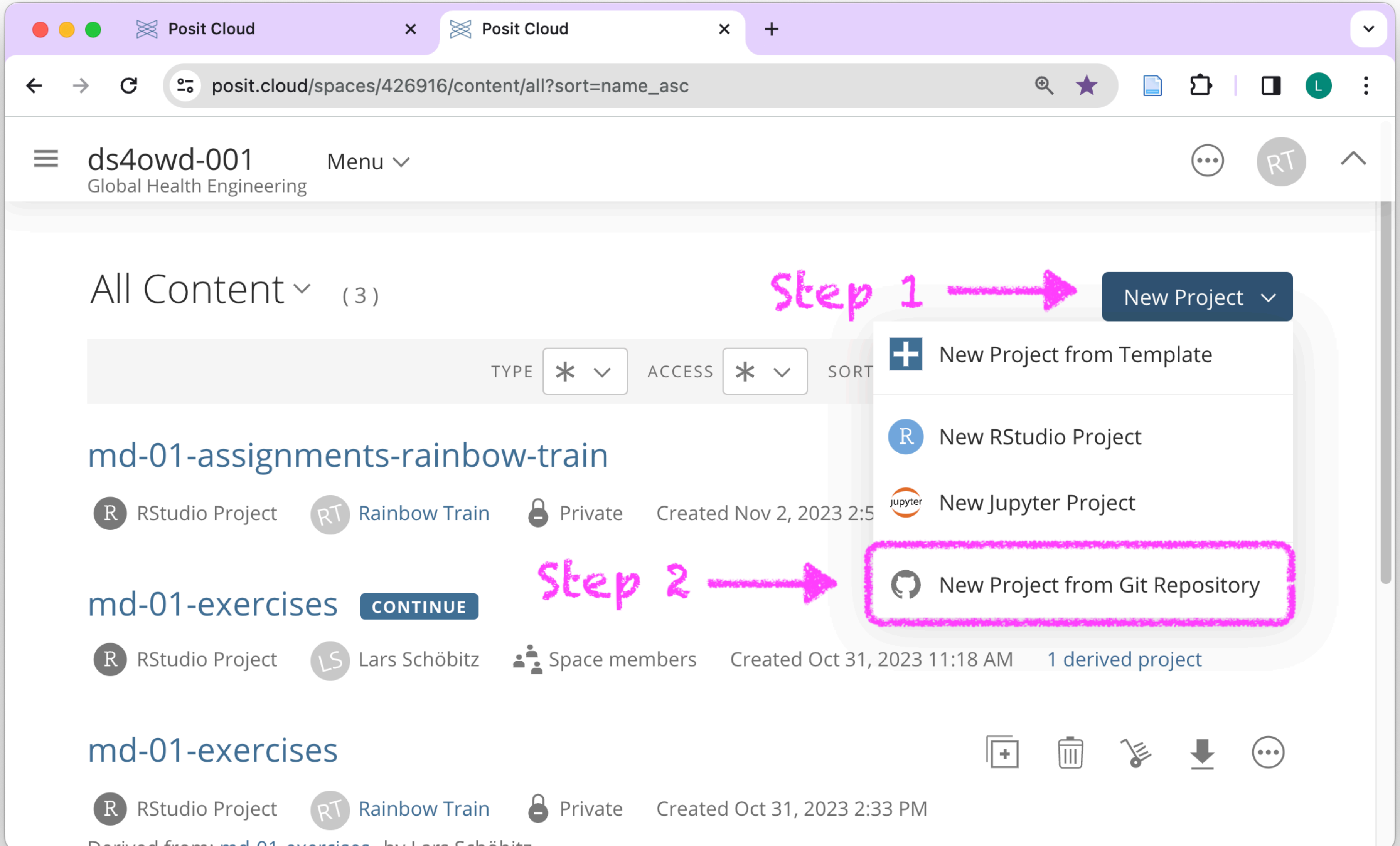
open wash data.

on your repository



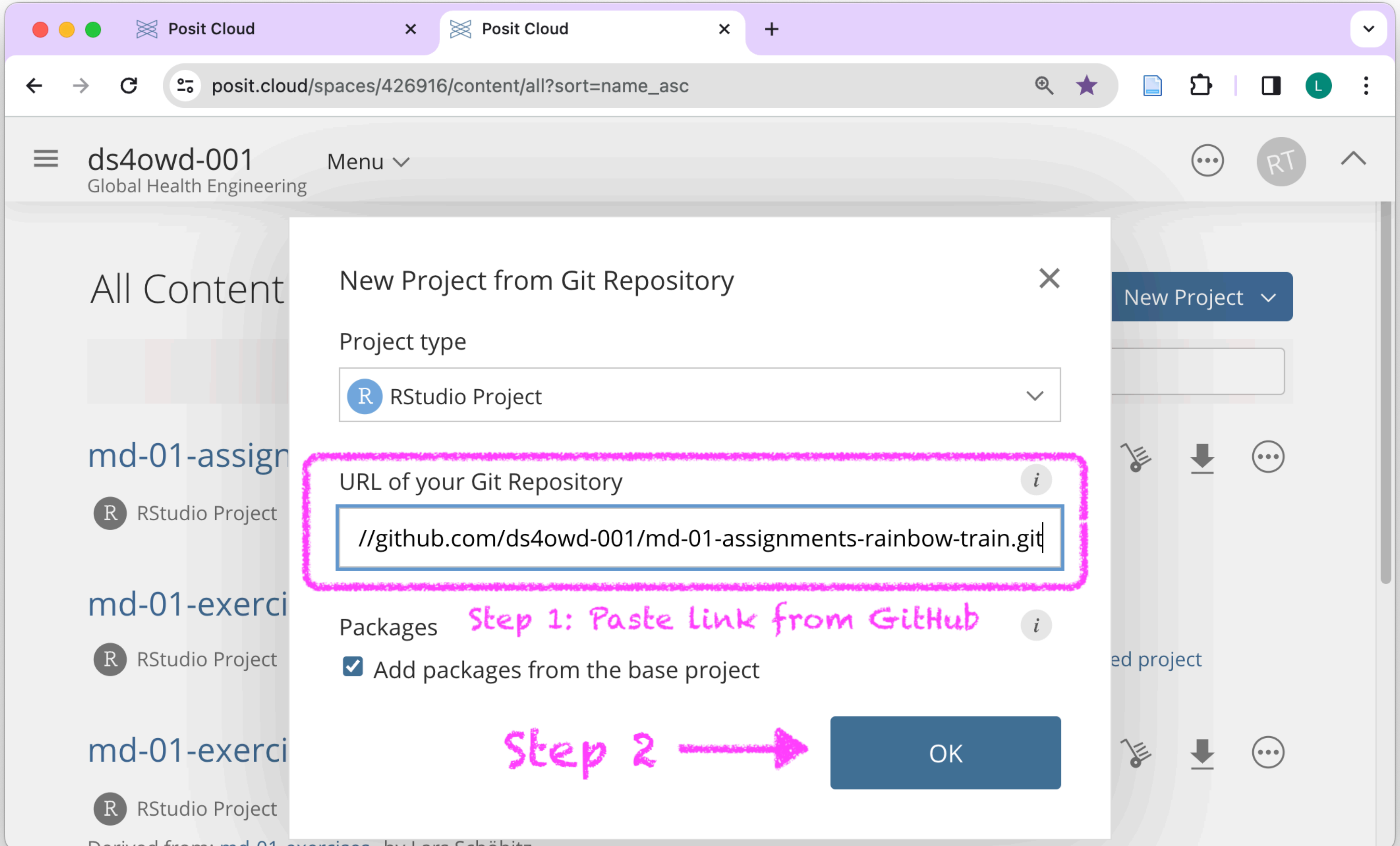
on Posit Cloud

posit.cloud/spaces/663318/



ds4owd-002.github.io/website/

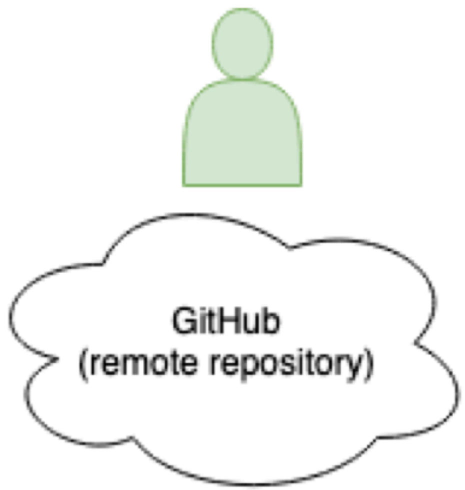
on Posit Cloud

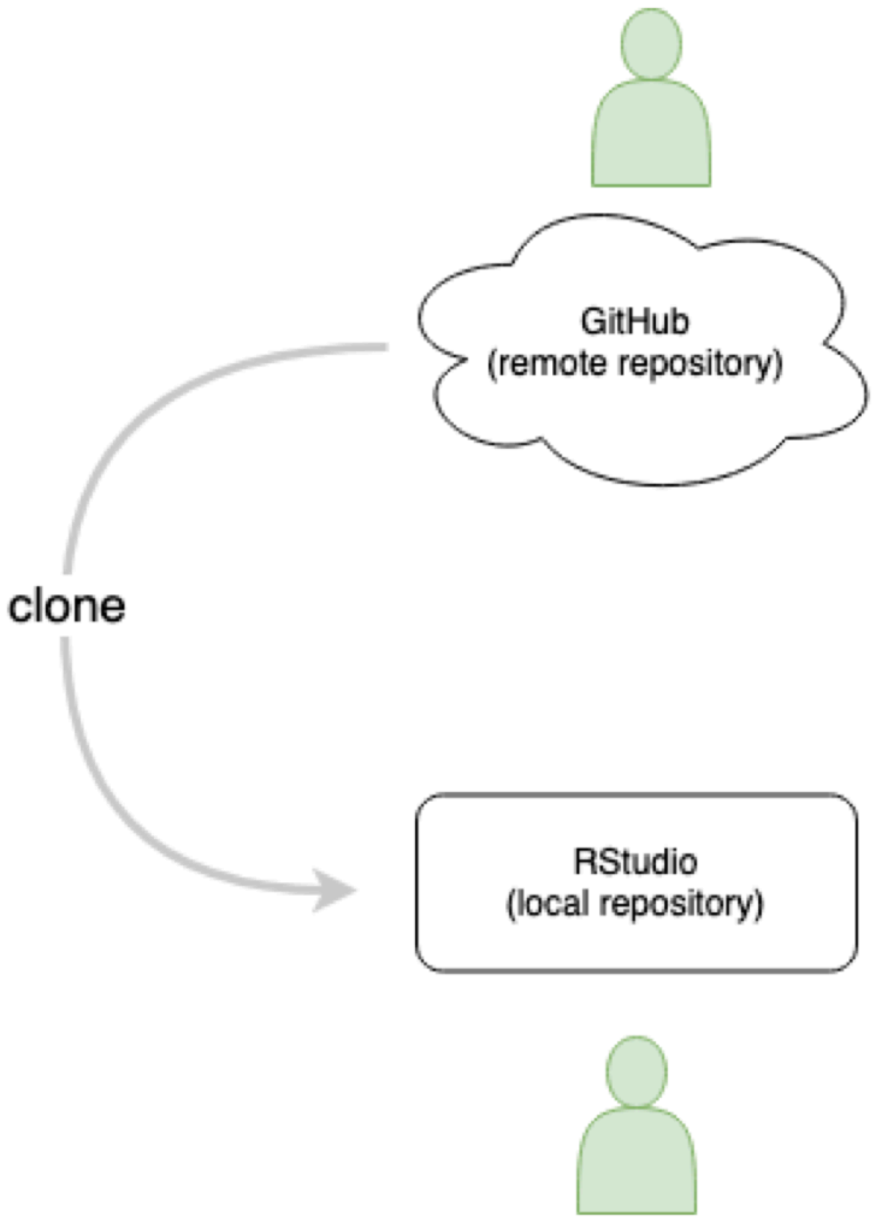

Posit Cloud
Posit Cloud
posit.cloud/spaces/426916/content/all?sort=name_asc
ds4owd-001
Global Health Engineering
Menu
RT
All Content
New Project
md-01-assign
RStudio Project
md-01-exerci
RStudio Project
ed project
md-01-exerci
RStudio Project
Derived from: md-01-exercises by Lars Schöbitz
New Project from Git Repository
Project type
RStudio Project
URL of your Git Repository
//github.com/ds4owd-001/md-01-assignments-rainbow-train.git
Packages
Step 1: Paste link from GitHub
Add packages from the base project
Step 2
OK

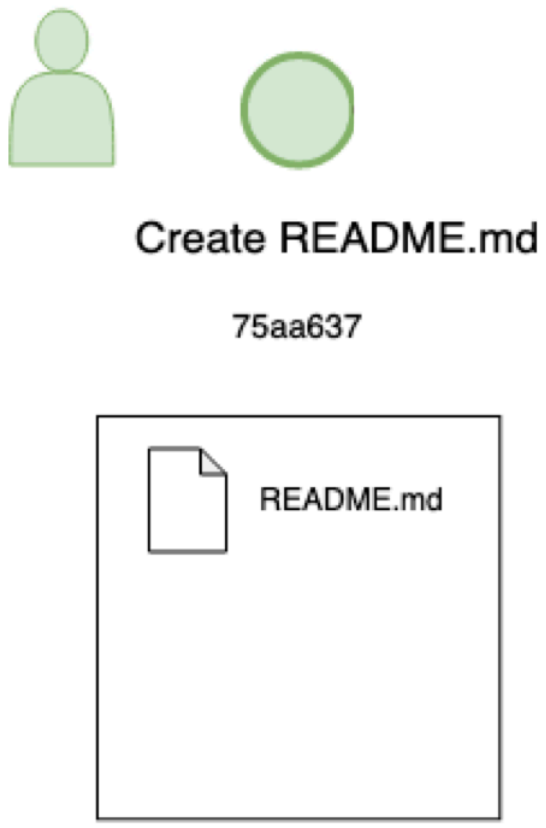
Module 2 - Assignment 1: Bookmarks

This week, you will have an assignment for creating bookmarks in your browser. It is very useful and will support you through the course and beyond.

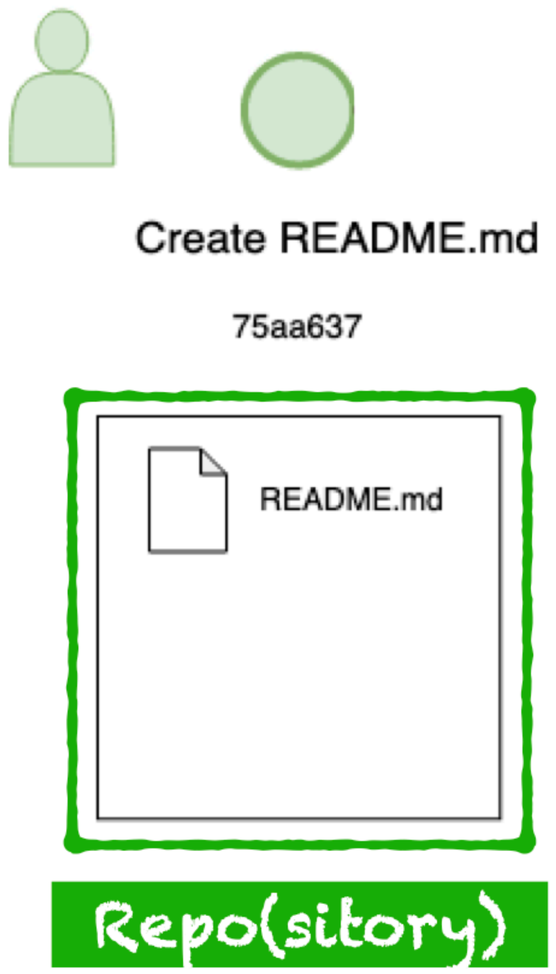
Version Control - Terminology

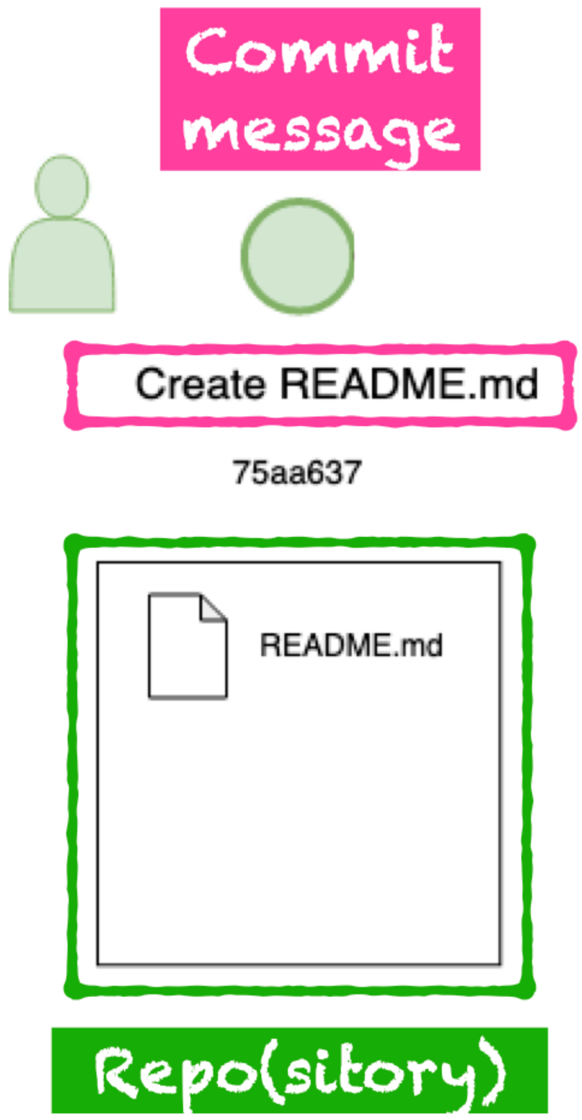

GitHub
(remote repository)

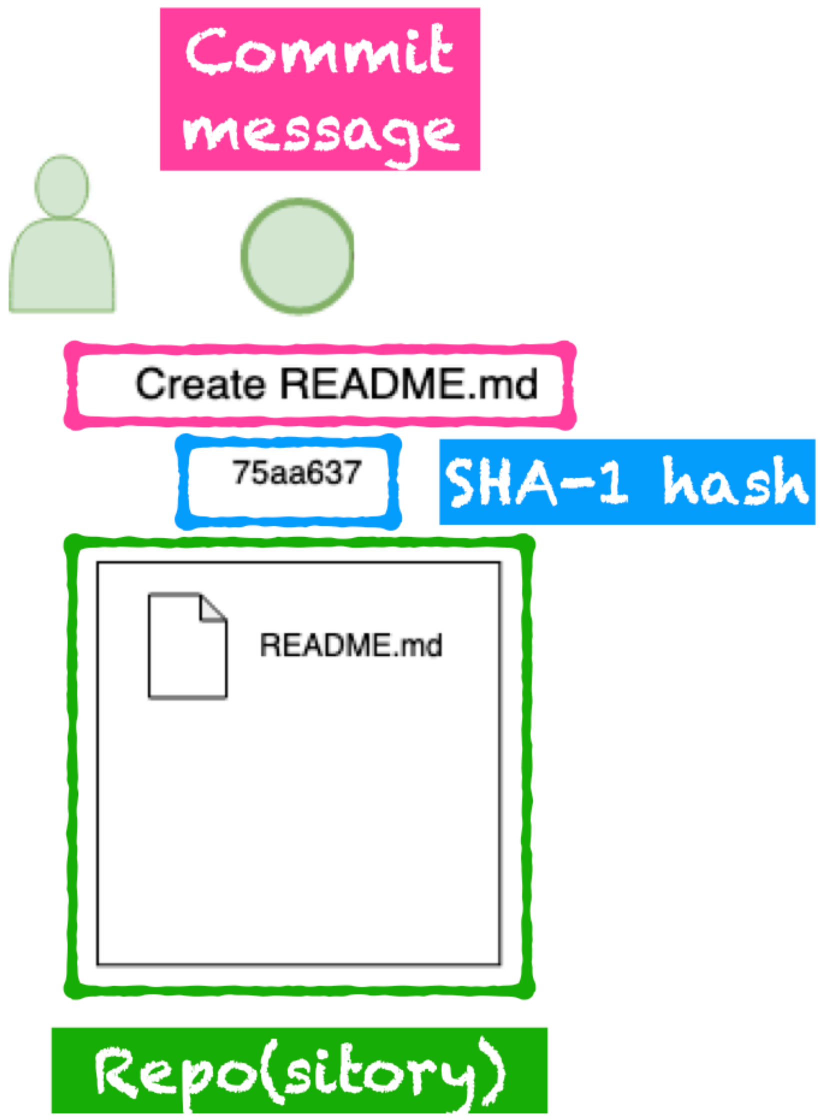

GitHub
(remote repository)
clone
RStudio
(local repository)

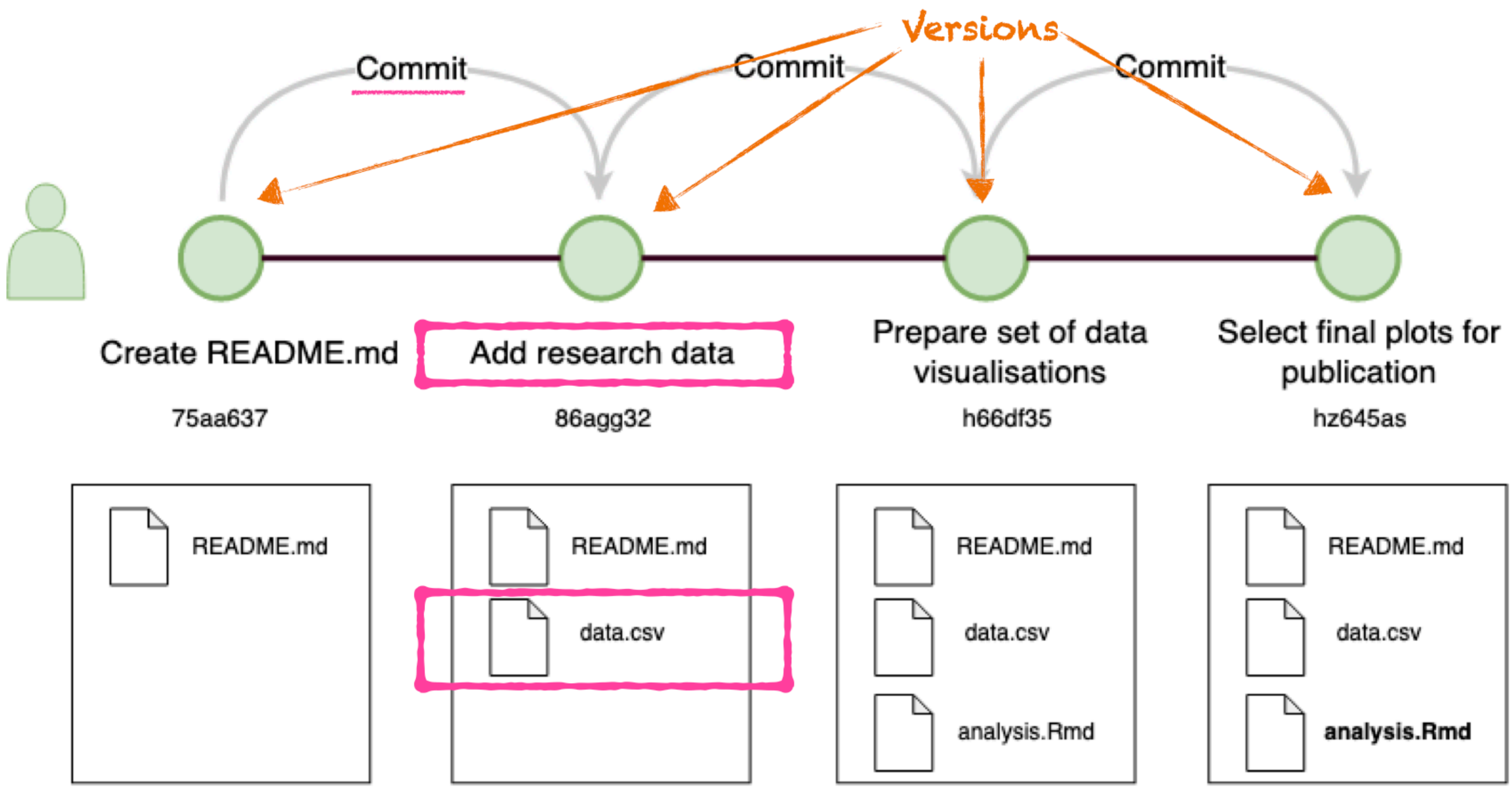


open
wash
data.


Create README.md
75aa637
README.md
Repo(sitory)


Commit message
Create README.md
75aa637
README.md
Repo(sitory)


Commit message
Create README.md
75aa637
SHA-1 hash
README.md
Repo(sitory)



Versions
Commit
Commit
Commit
Create README.md
Add research data
Prepare set of data visualisations
Select final plots for publication
75aa637
86agg32
h66df35
hz645as
README.md
README.md
data.csv
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analysis.Rmd
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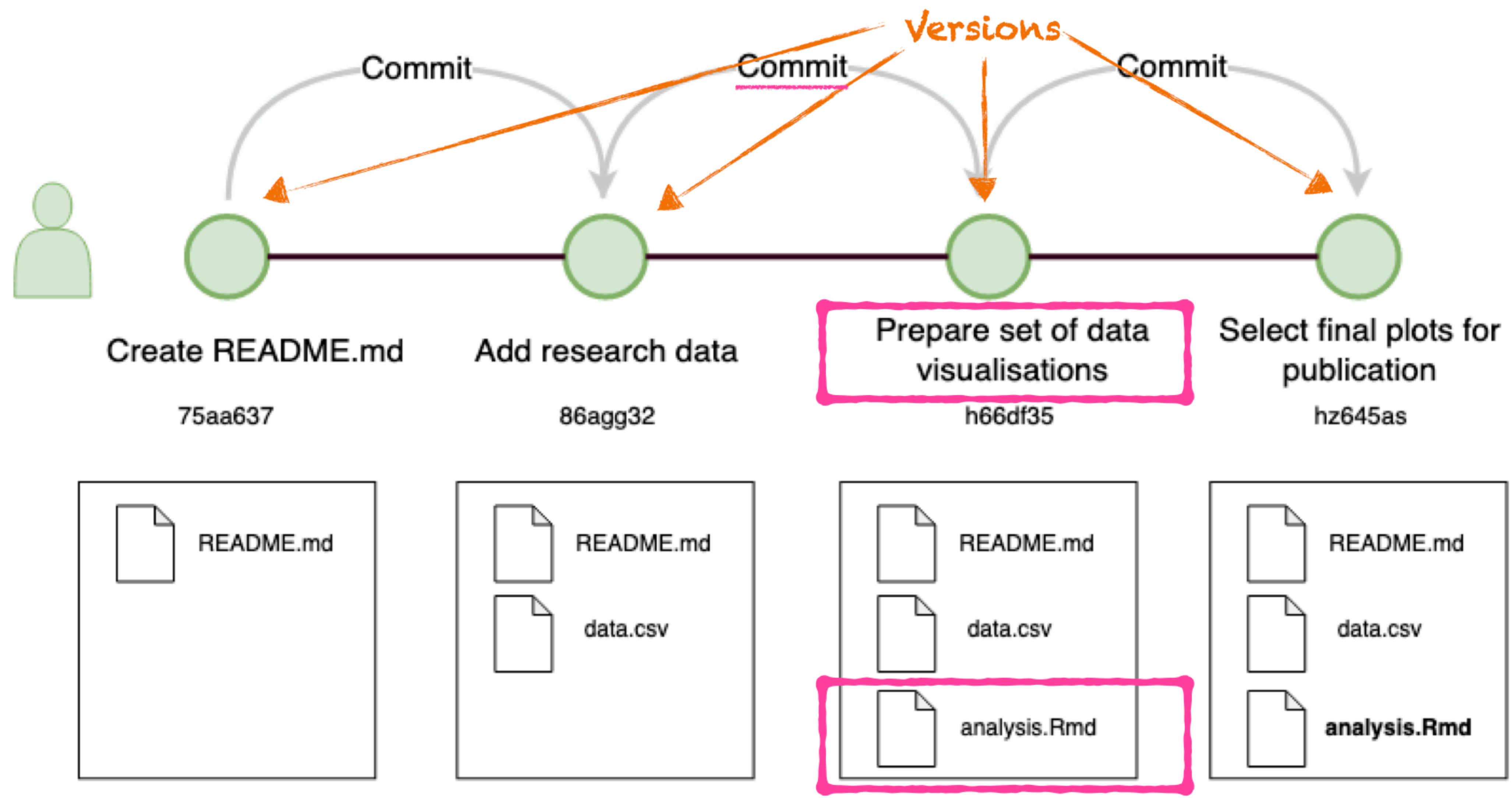

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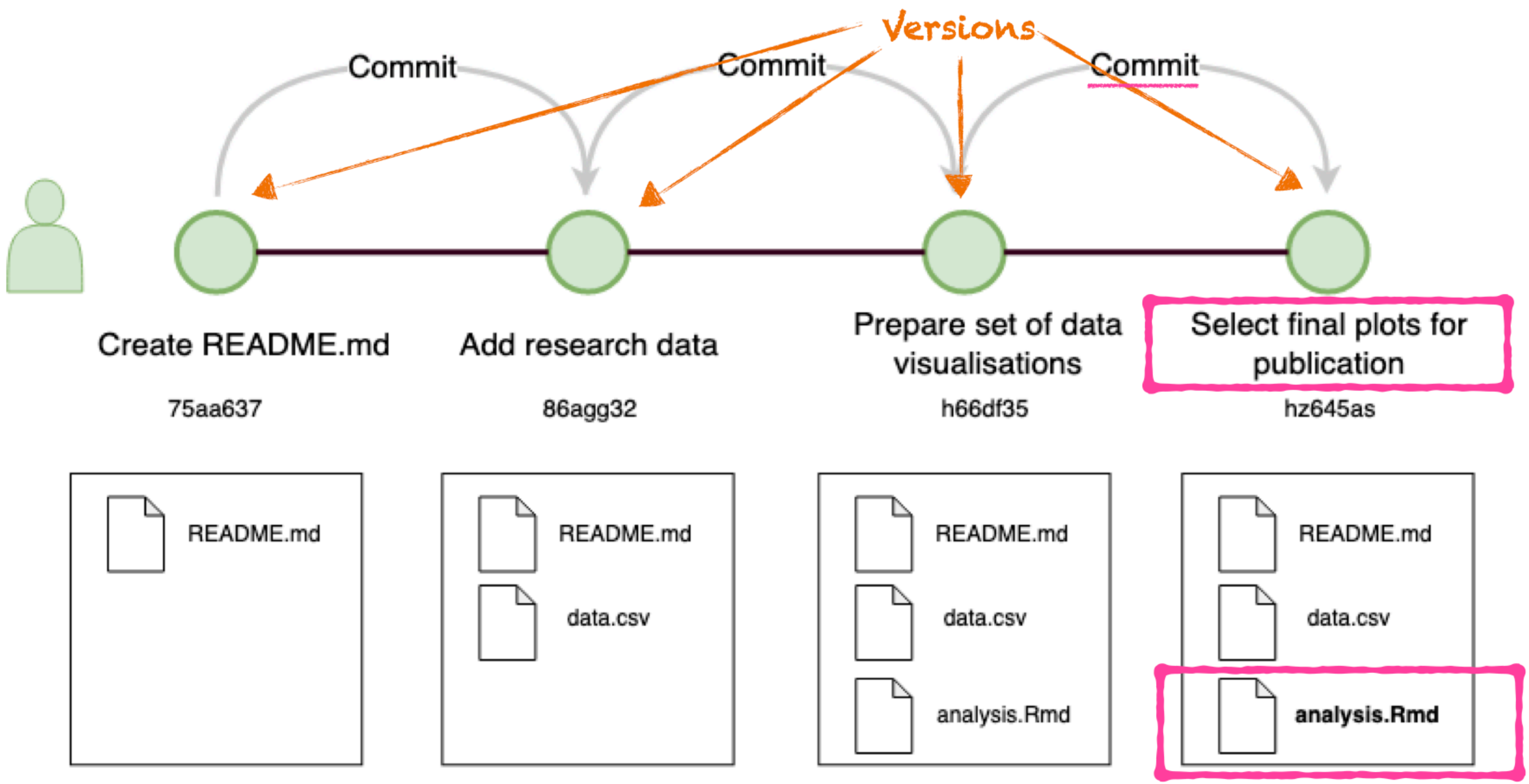
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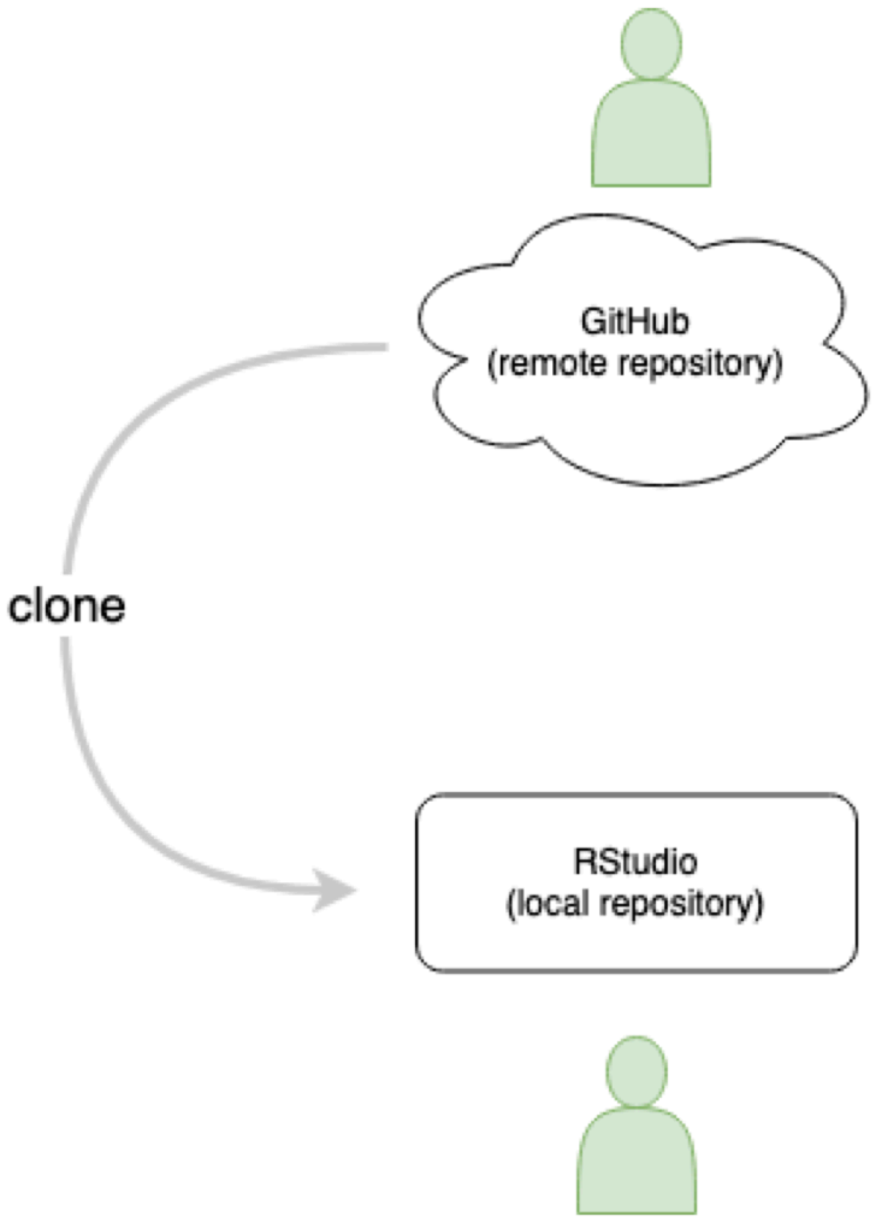
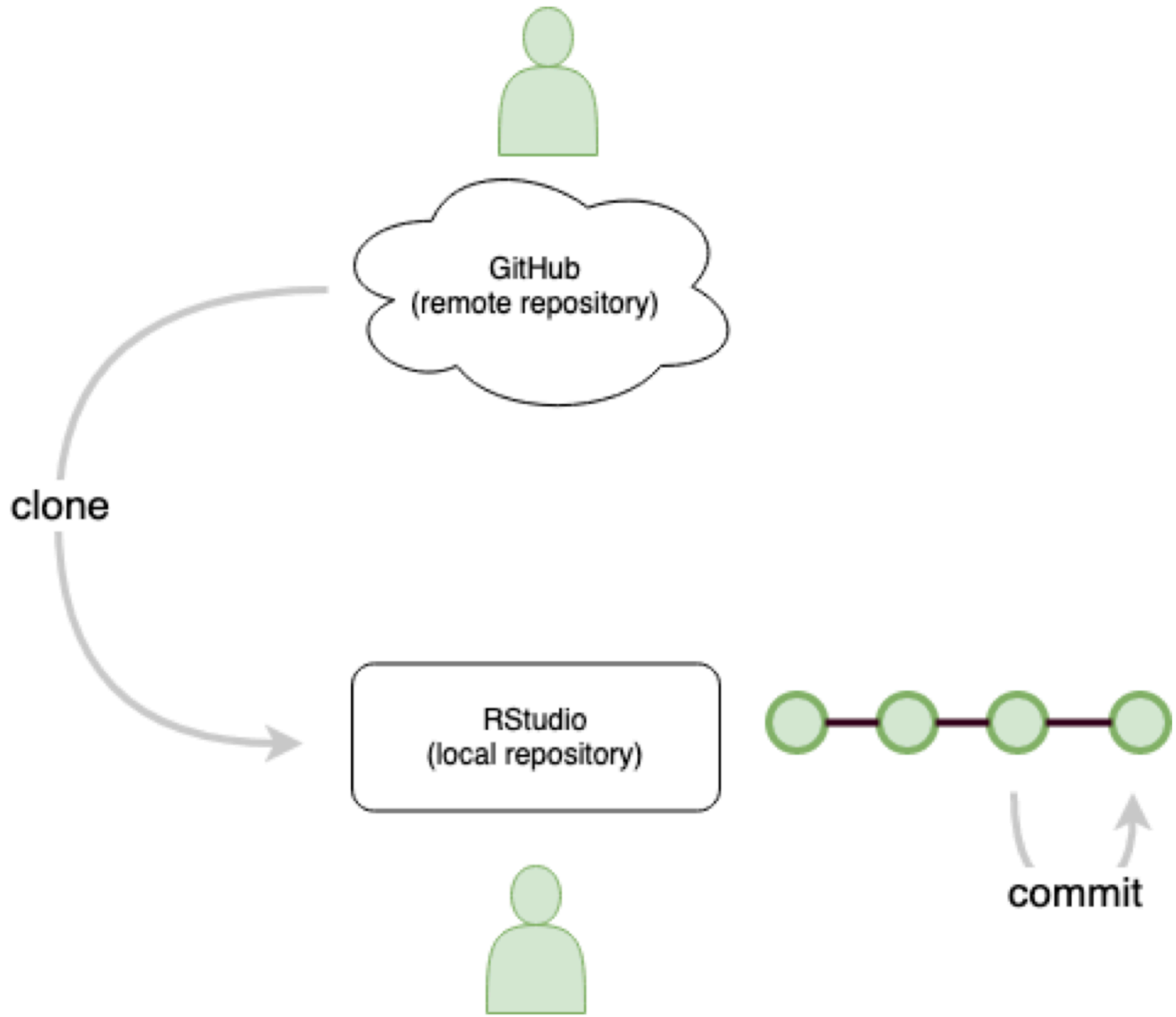
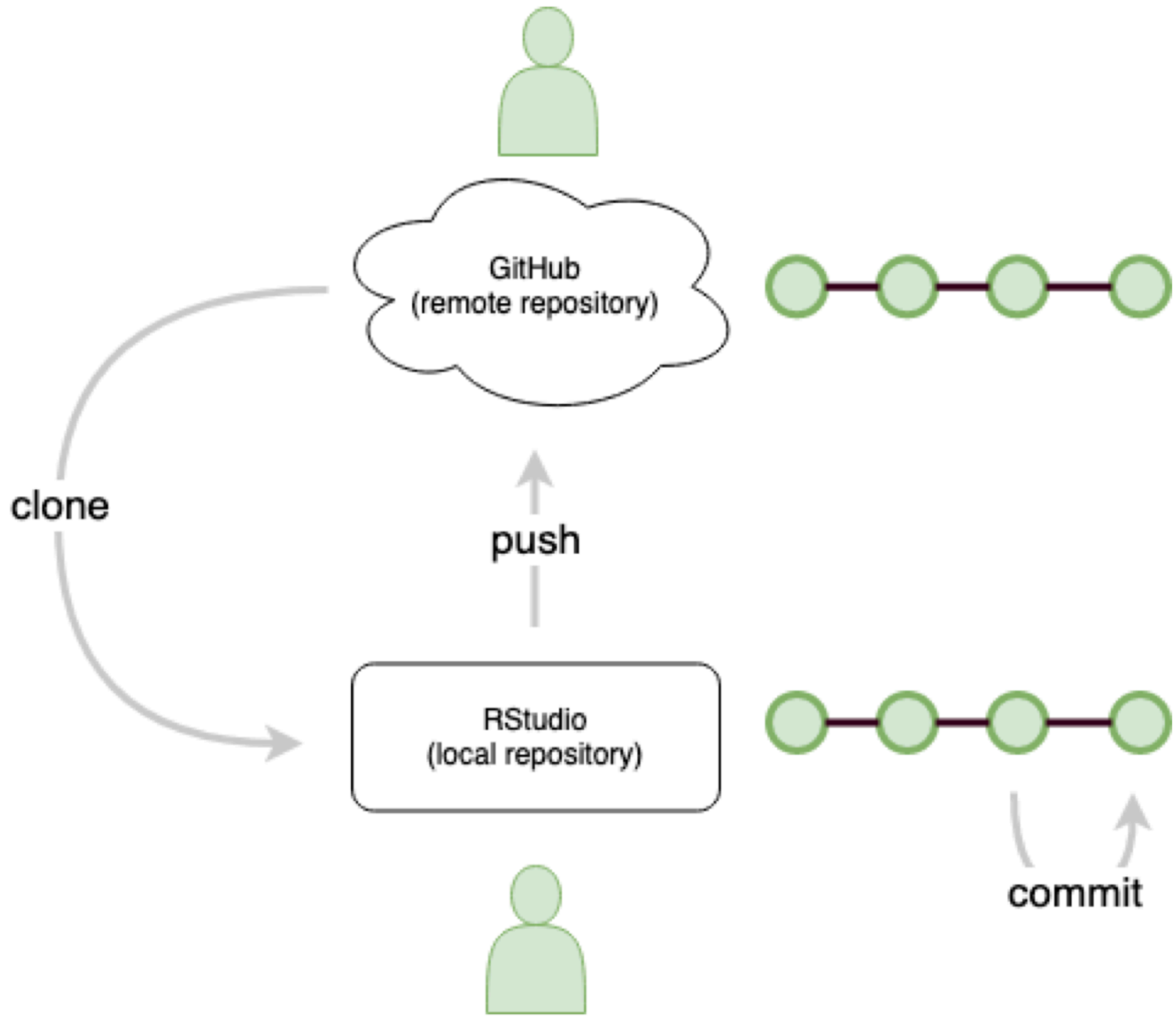
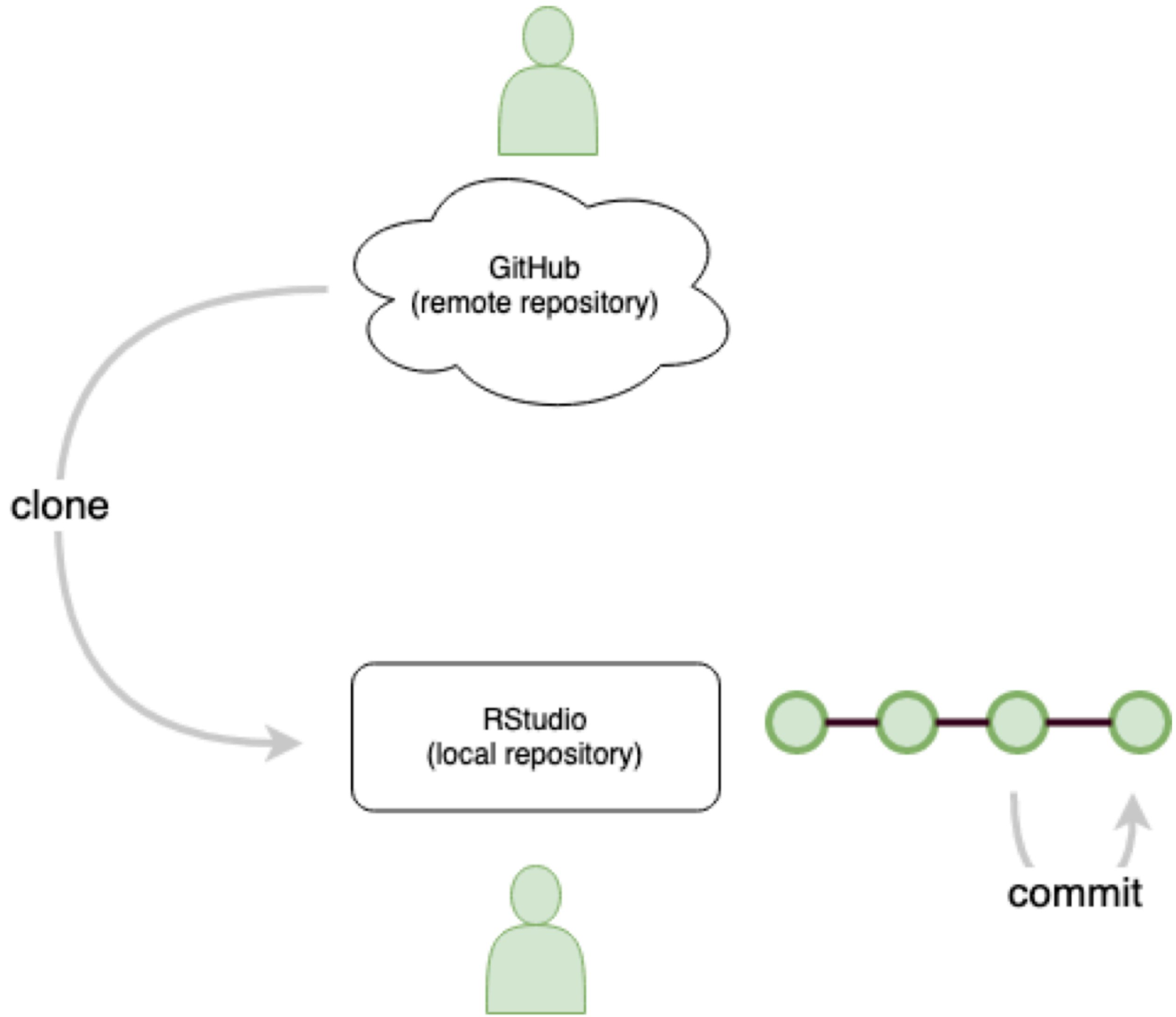
data.csv
analysis.Rmd


GitHub
(remote repository)
clone
RStudio
(local repository)

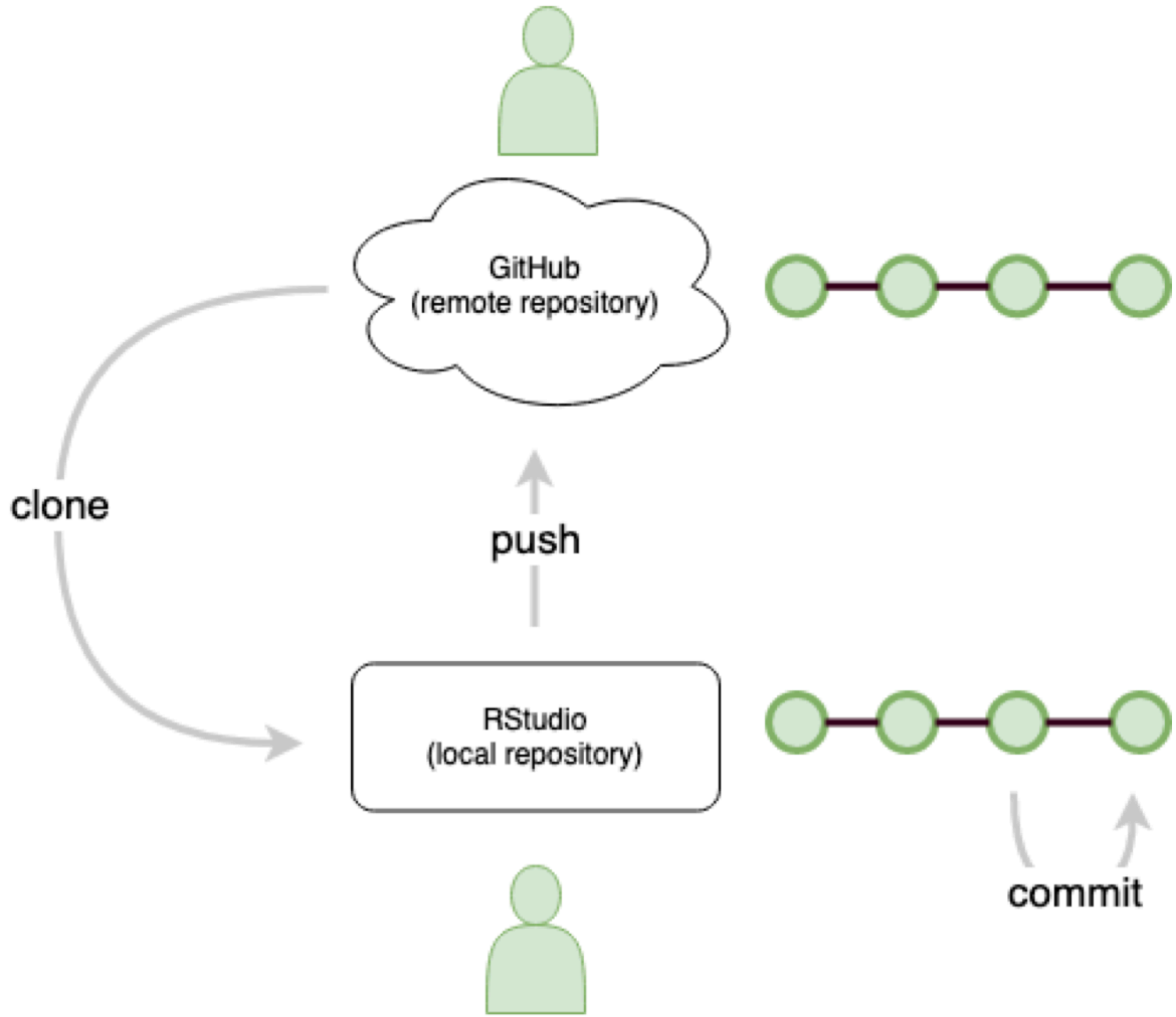

GitHub
(remote repository)
clone
RStudio
(local repository)
commit


GitHub
(remote repository)
clone
push
RStudio
(local repository)
commit

remember: `git commit`

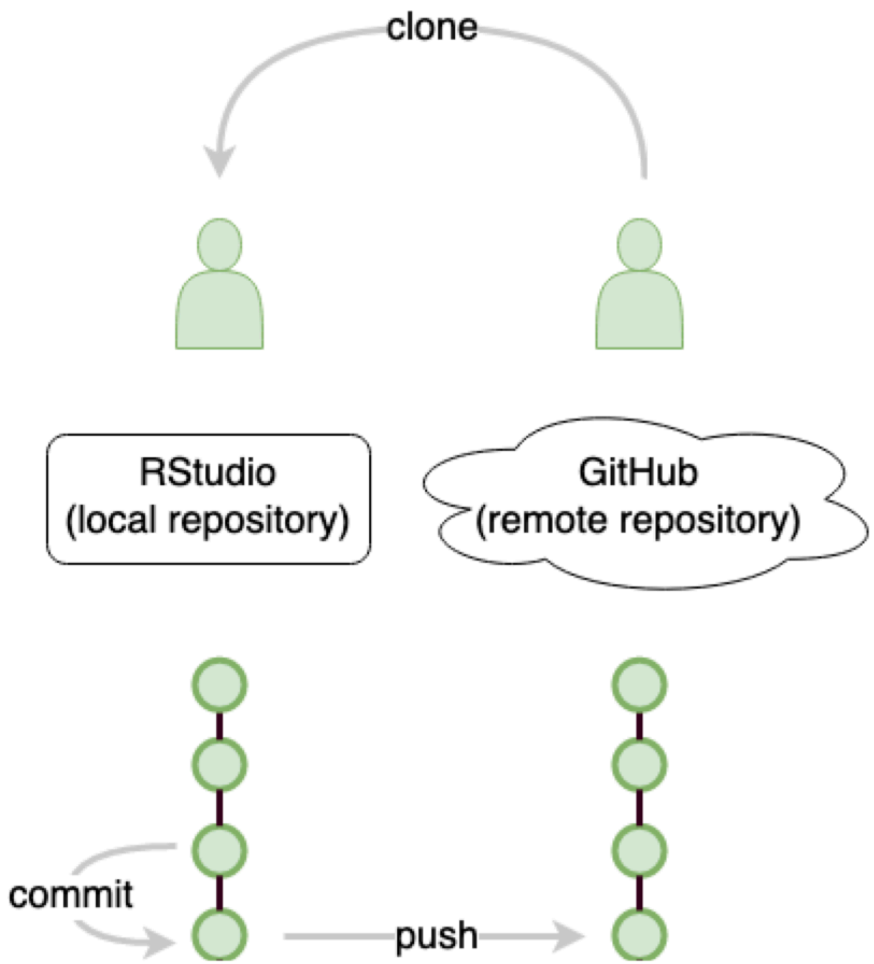


remember: `git push`

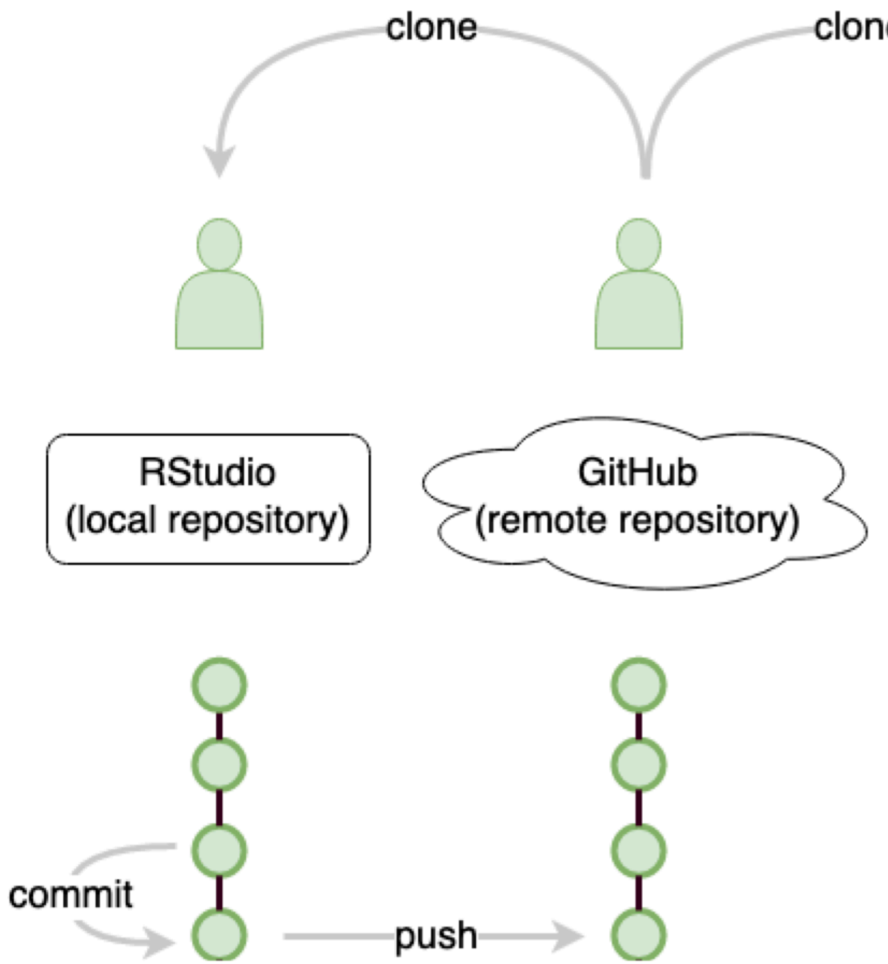


open
wash
data.

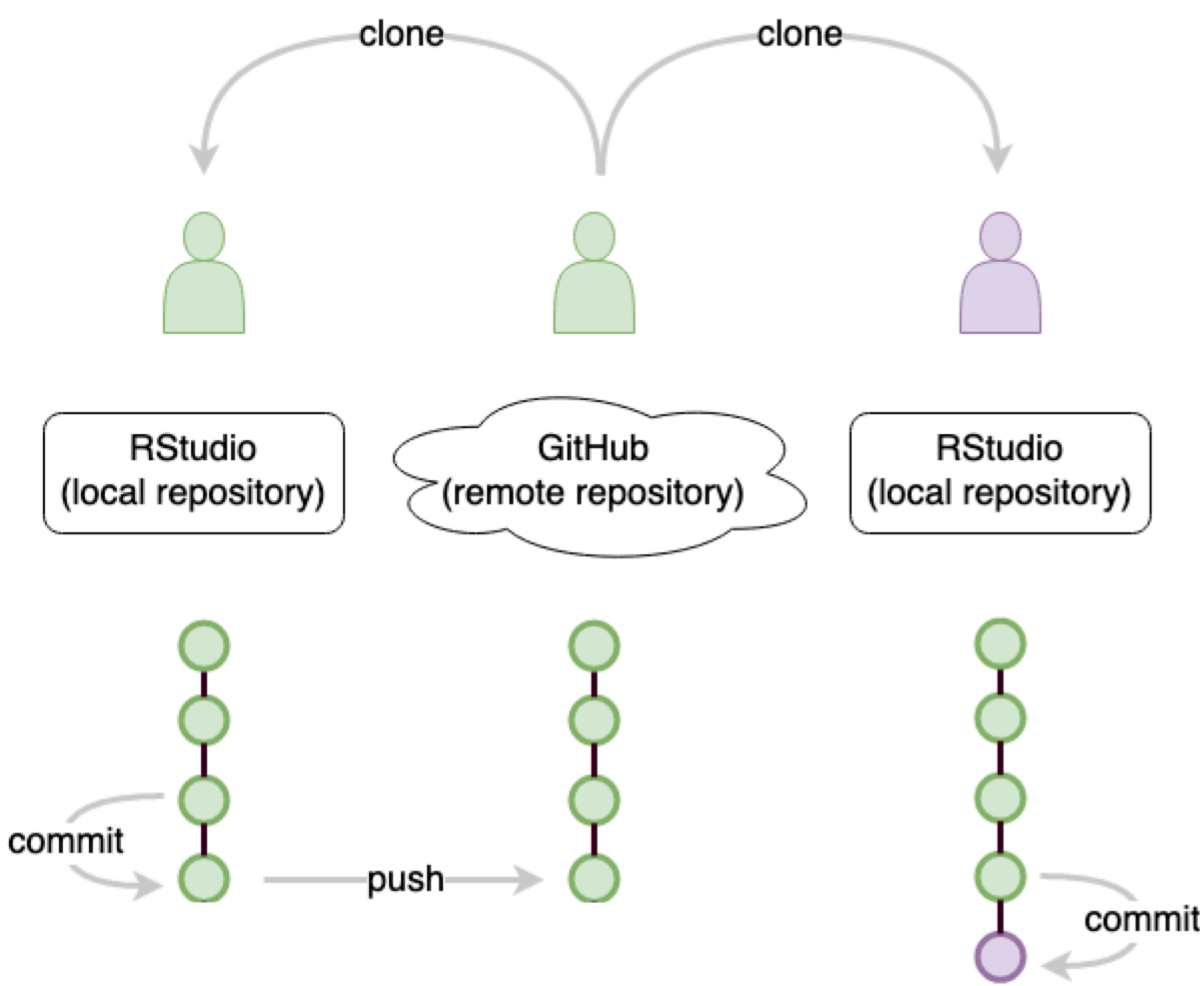
remember: `git push`



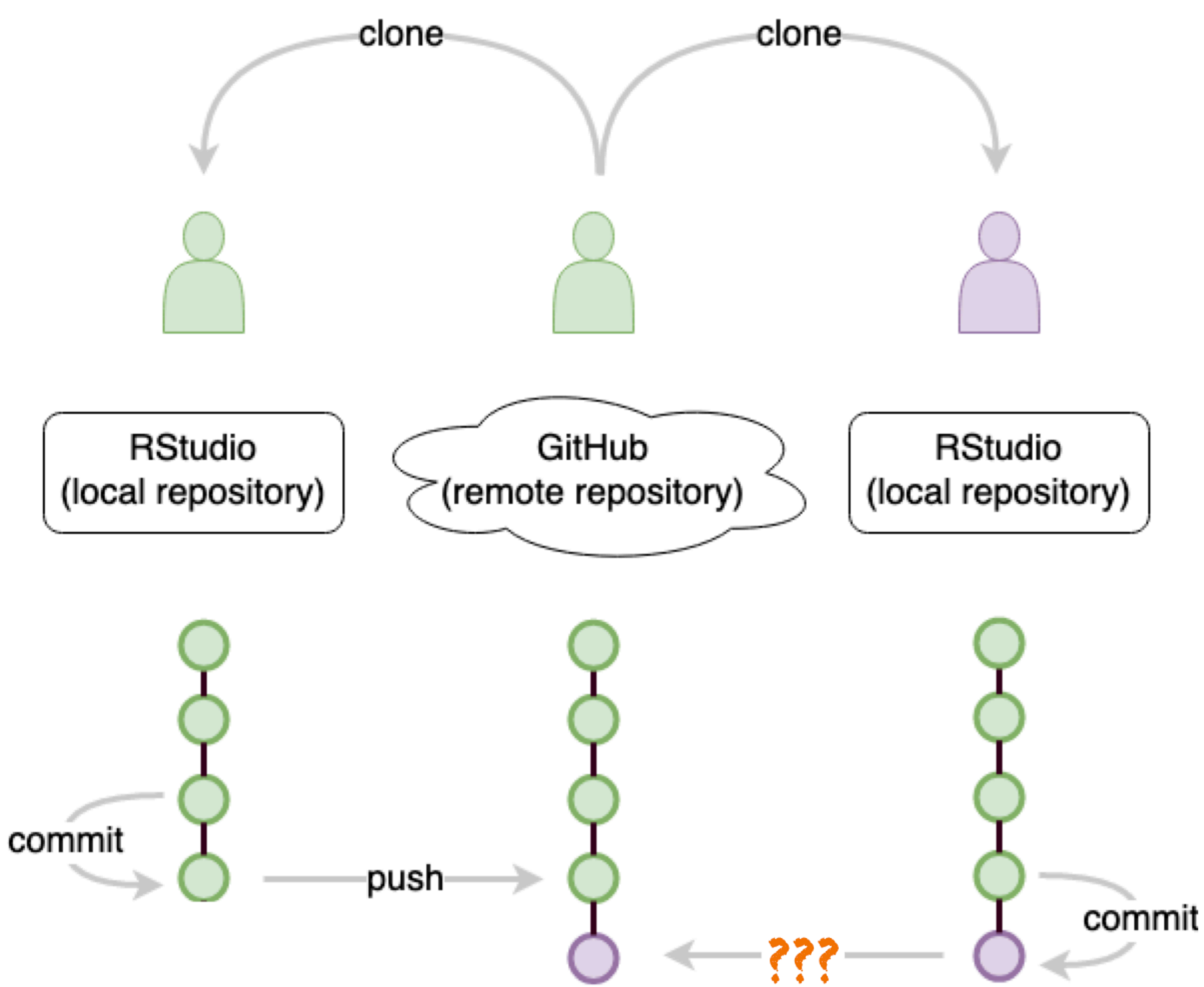
collaborate: `git clone`



track work: `git commit`

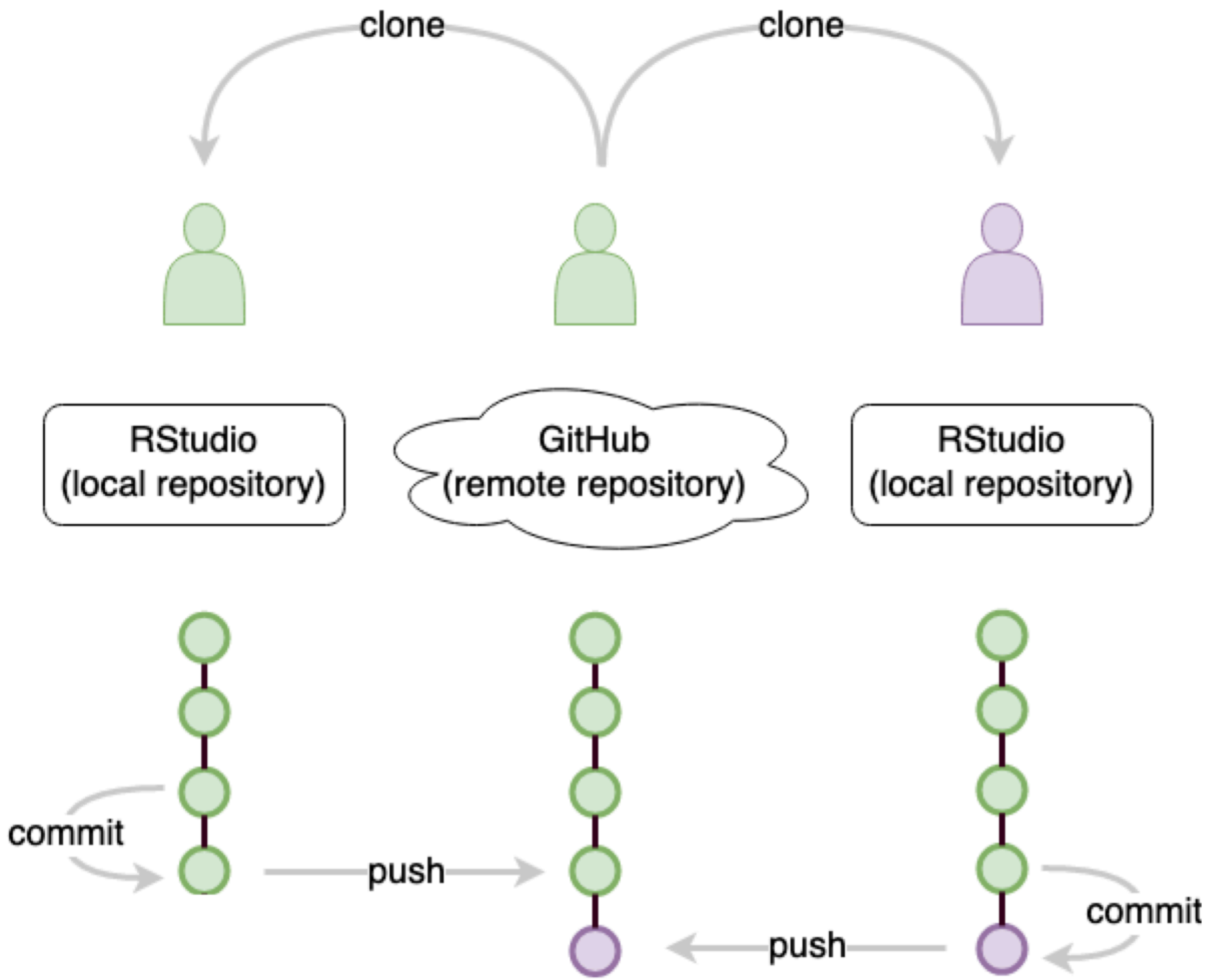


update: `git ???`

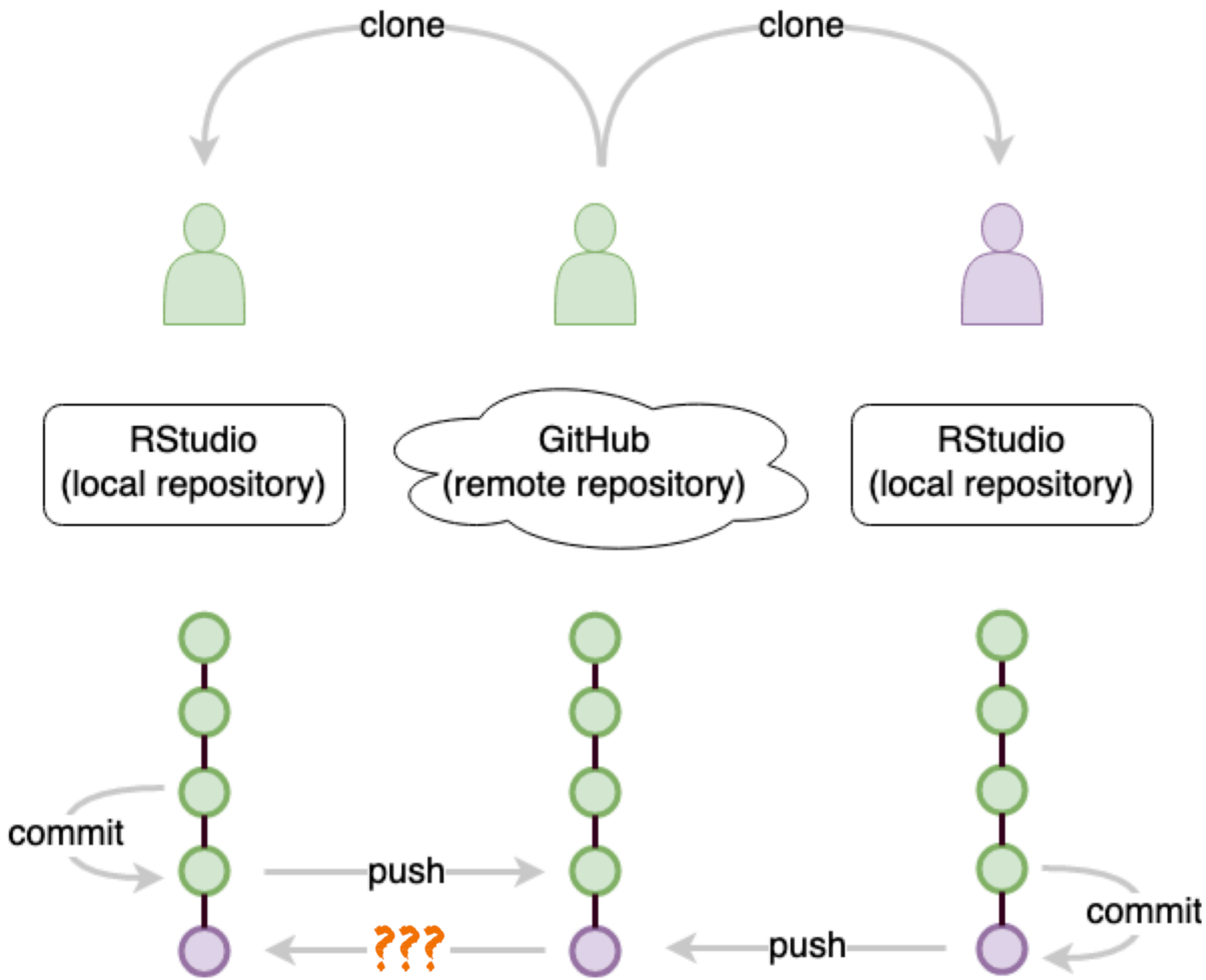


open
wash
data.

update: `git push`

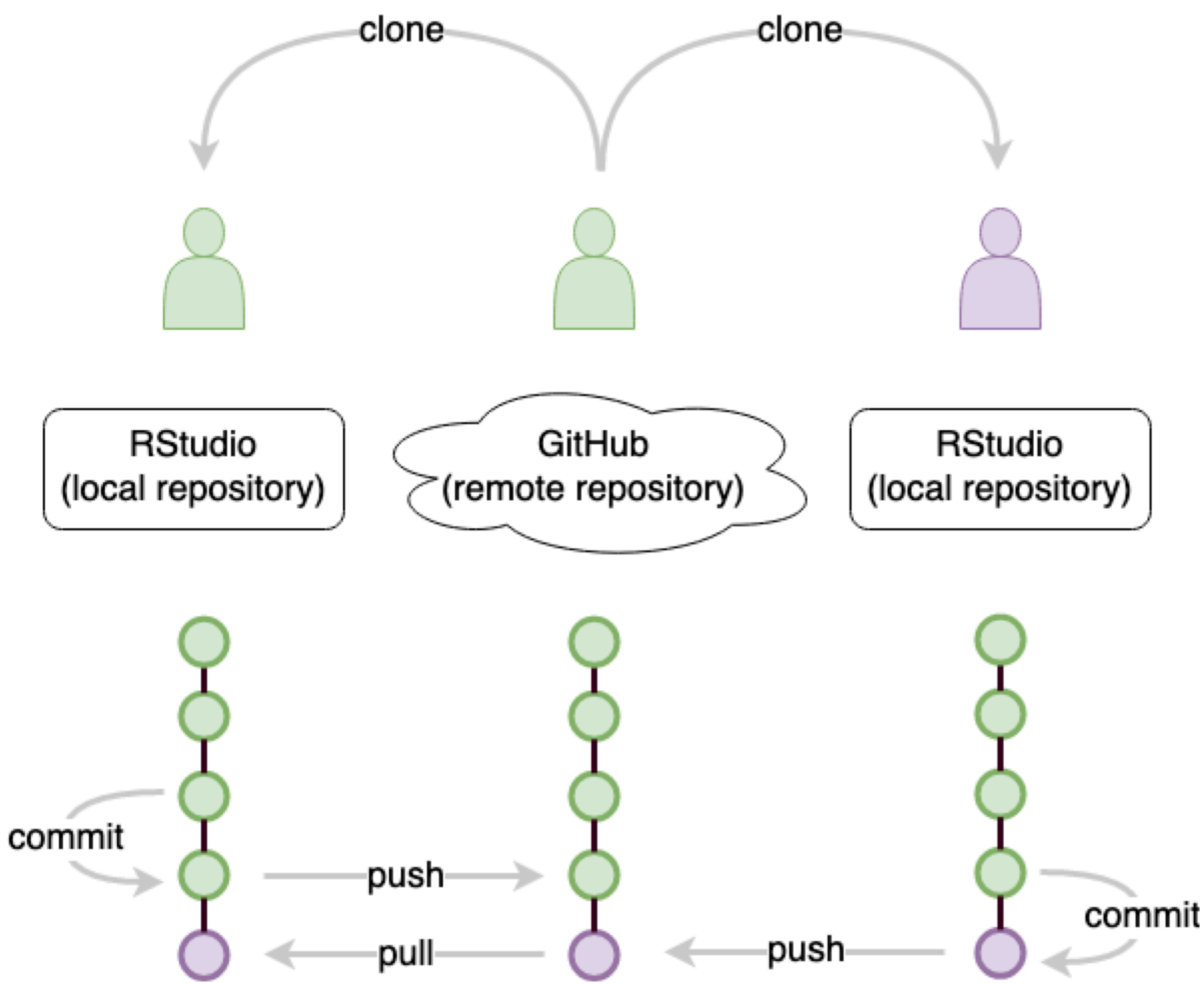


git ???



clone
clone
RStudio
(local repository)
GitHub
(remote repository)
RStudio
(local repository)
commit
push
???
push
commit

new: `git pull`

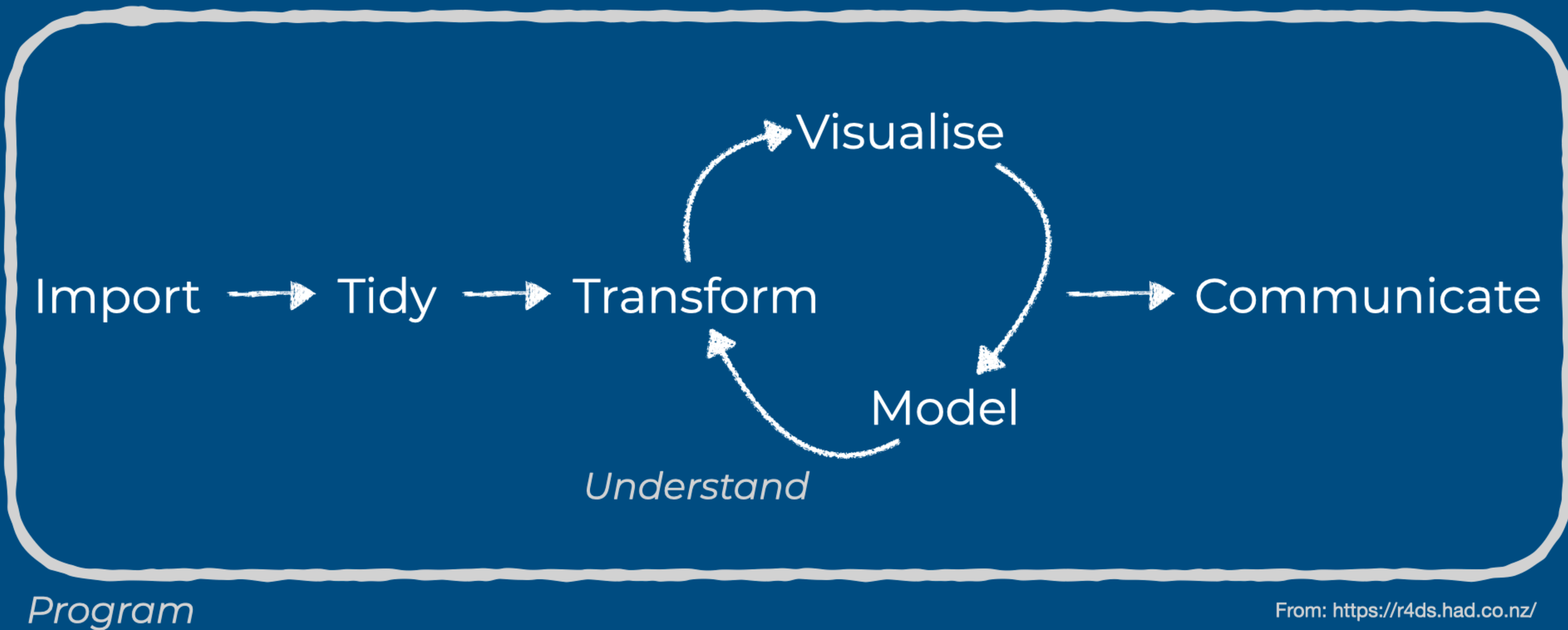

clone
clone
RStudio
(local repository)
GitHub
(remote repository)
RStudio
(local repository)
commit
push
pull
push
commit

Learning Objectives (for this week)

1. Learners can list the six elements of the data science lifecycle.
2. Learners can describe the four main aesthetic mappings that can be used to visualise data using the ggplot2 R Package.
3. Learners can control the colour scaling applied to a plot using colour as an aesthetic mapping.
4. Learners can compare three different geoms (bar/col, histogram, point) and their use case.

Data Science Lifecycle

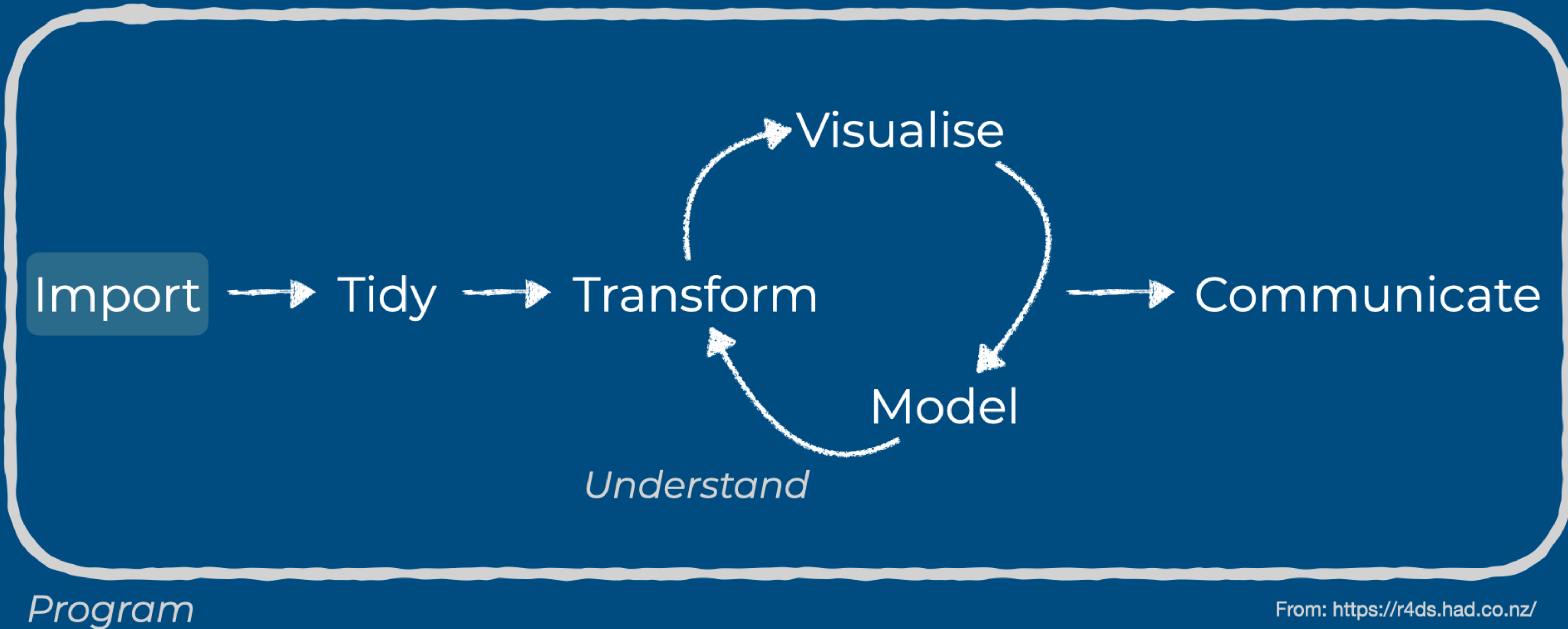
Data Science Lifecycle



From: https://r4ds.had.co.nz/

Data Science Lifecycle

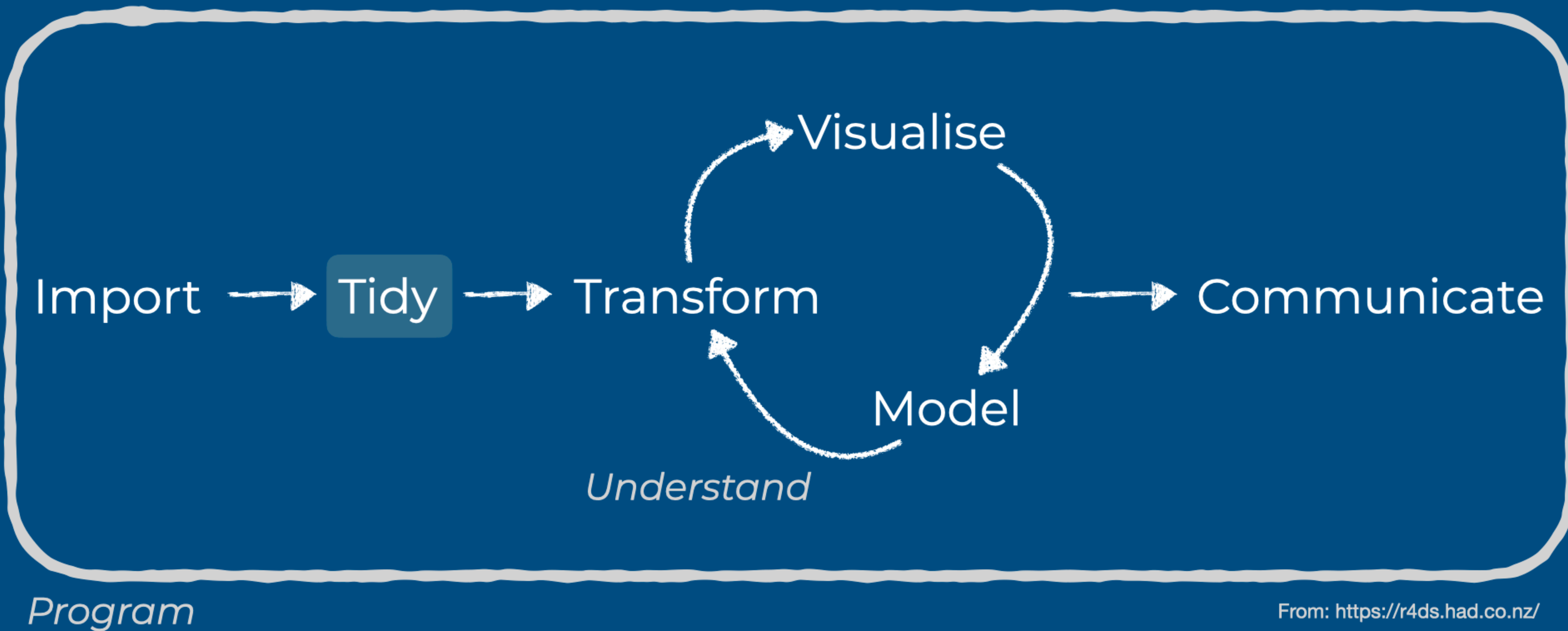
Get your data into R



From: https://r4ds.had.co.nz/

Data Science Lifecycle

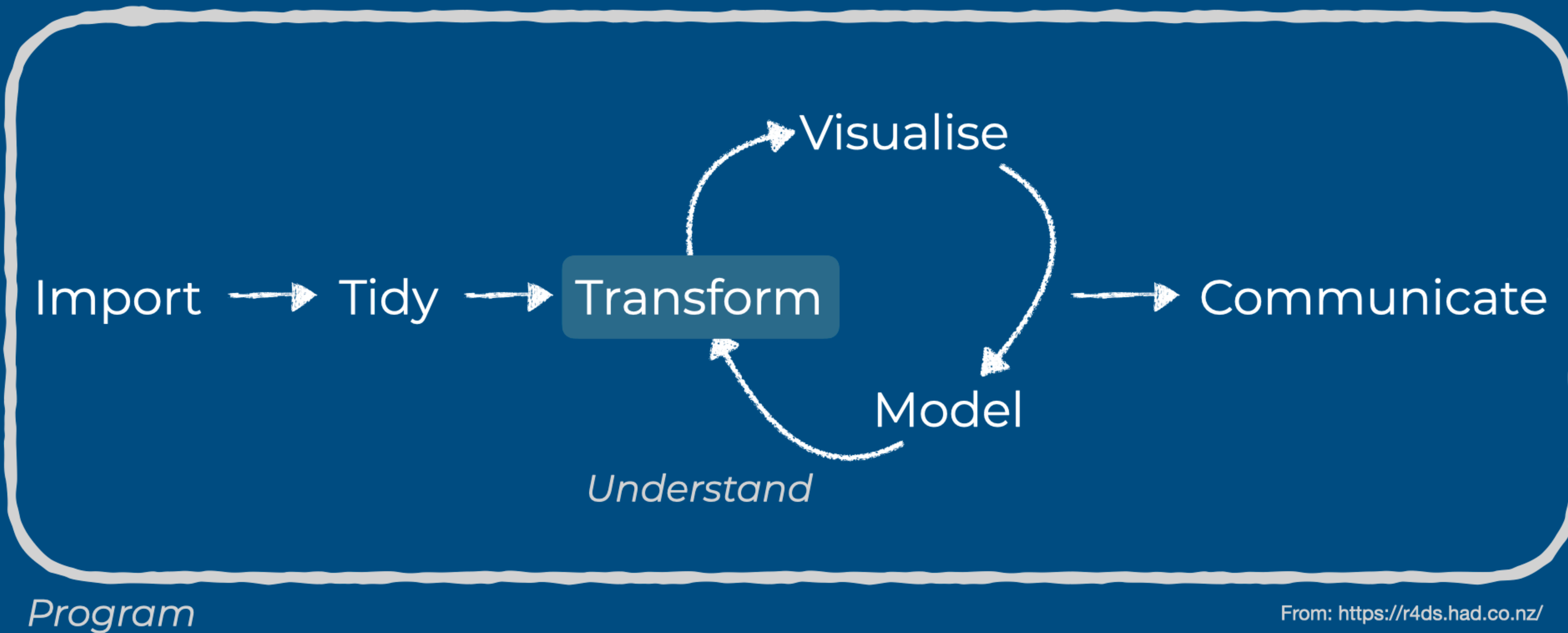
Store your data in a consistent form



From: https://r4ds.had.co.nz/

Data Science Lifecycle

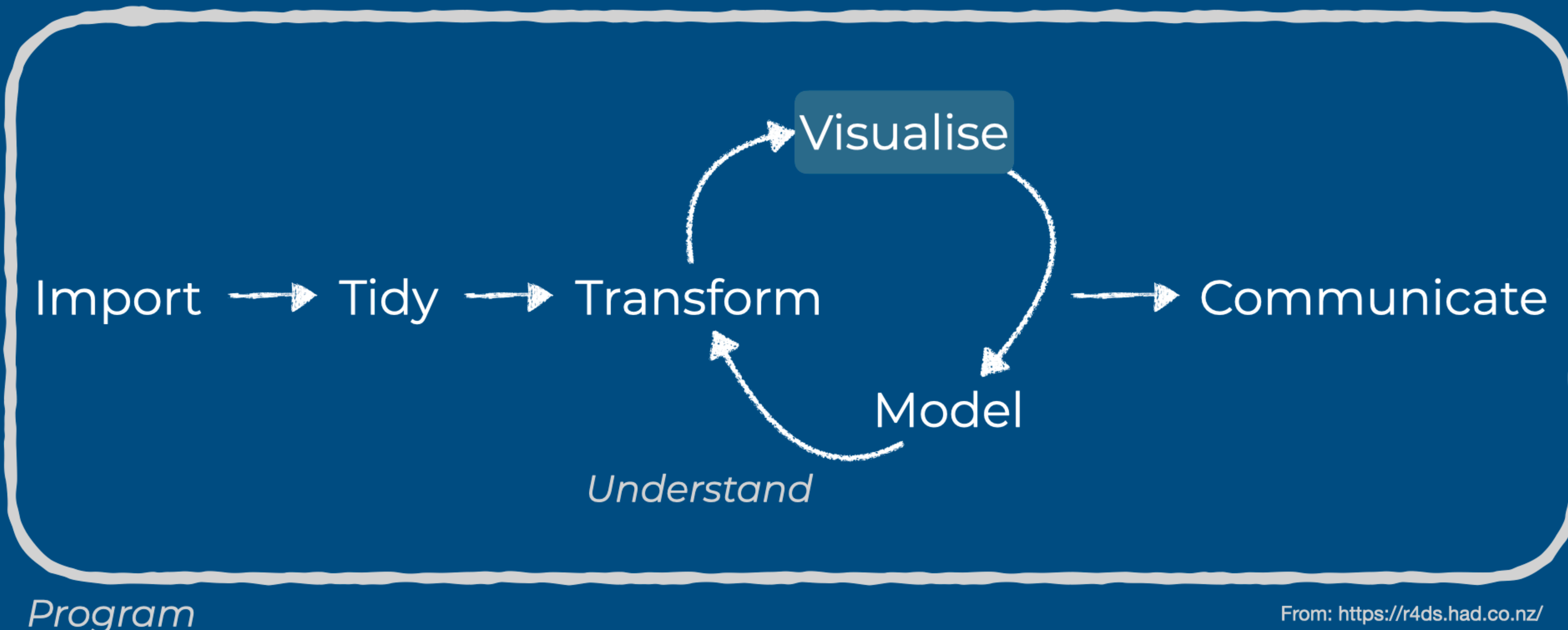
Narrow down + Create new variables + Summary stats



From: https://r4ds.had.co.nz/

Data Science Lifecycle

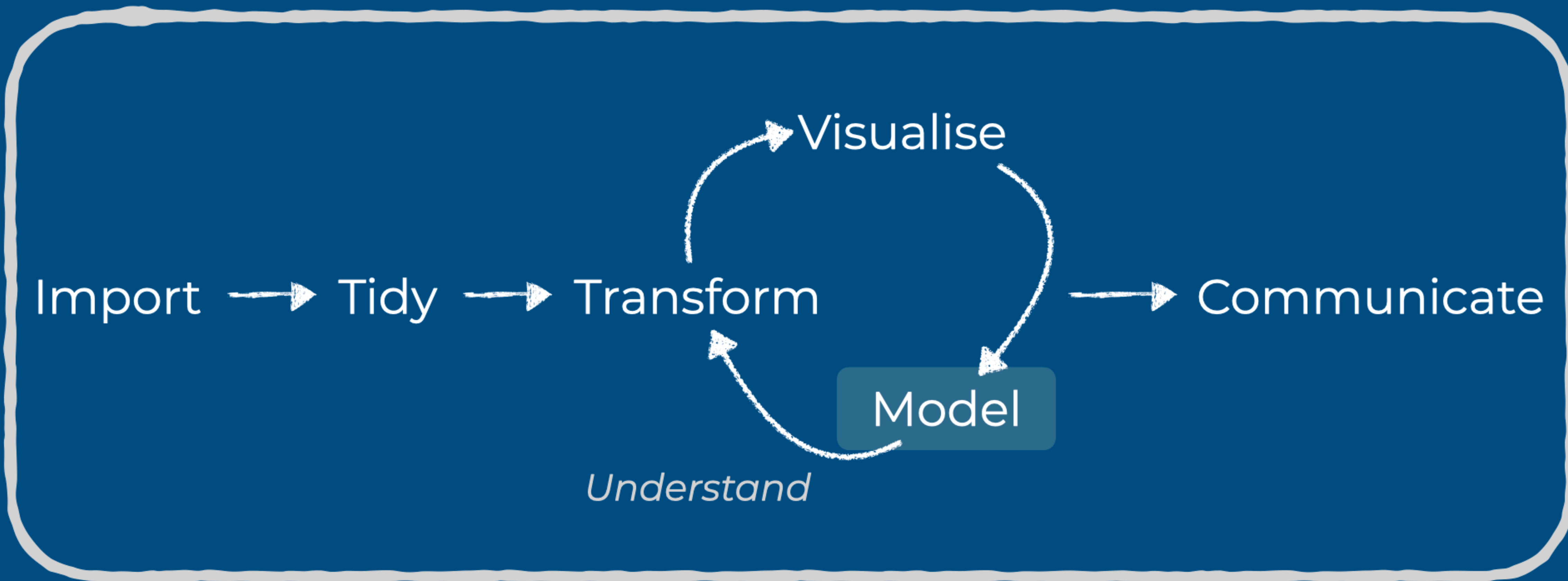
Explore your with visual representations



From: https://r4ds.had.co.nz/

Data Science Lifecycle

Explore your with visual representations

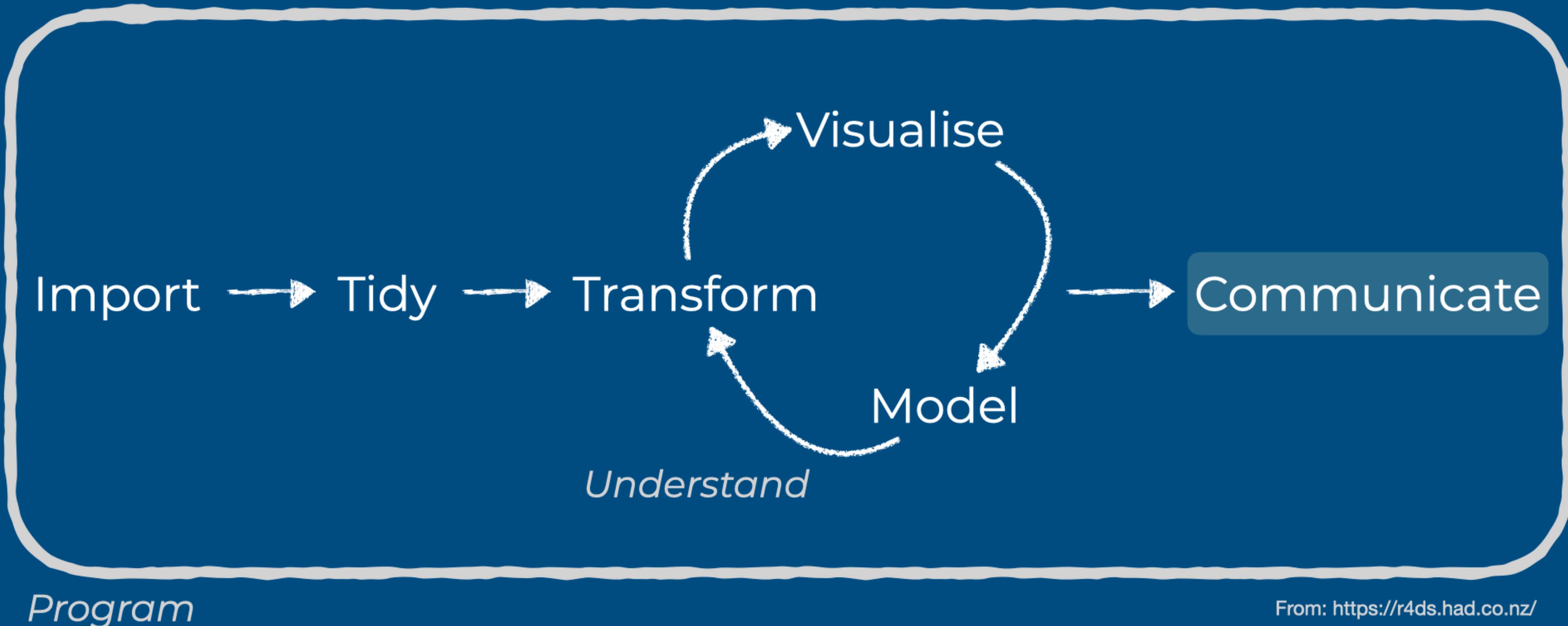


Program

From: https://r4ds.had.co.nz/

Data Science Lifecycle

Share your findings with others

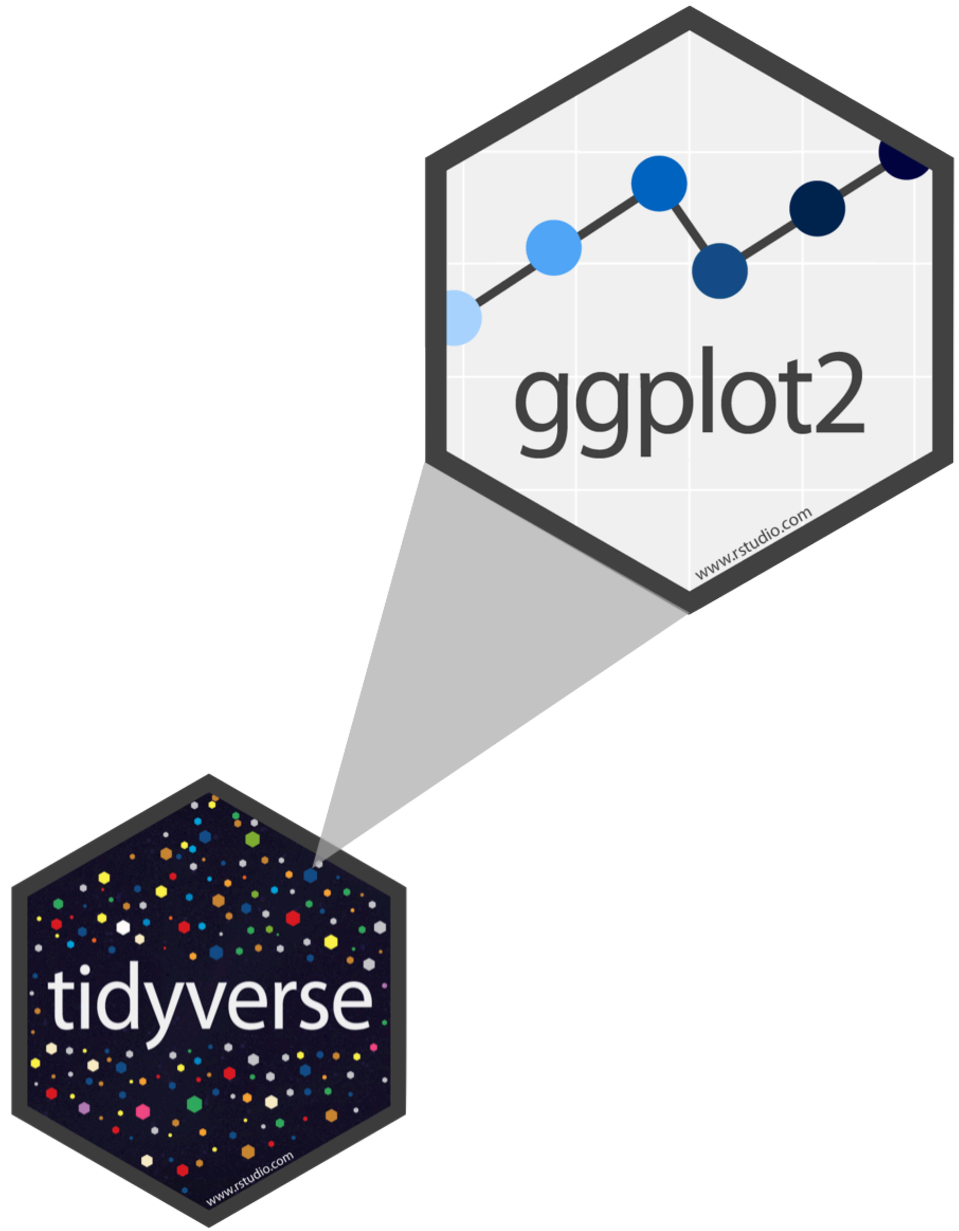


From: https://r4ds.had.co.nz/

Exploratory Data Analysis with `ggplot2`

R Package `ggplot2`

- **ggplot2** is tidyverse’s data visualization package
- `gg` in `ggplot2` stands for Grammar of Graphics
- Inspired by the book [Grammar of Graphics by Leland Wilkinson]
- **Documentation:** [https://ggplot2.tidyverse.org/]



My turn: Working with R

Sit back and enjoy!

Take a break

Please get up and move!

10:00

open wash data.

Code structure

- `ggplot()` is the main function in ggplot2
- Plots are constructed in layers
- Structure of the code for plots can be summarized as

```
ggplot(data = [dataset],
       mapping = aes(x = [x-variable],
                     y = [y-variable])) +
  geom_xxx() +
  other options
```

Code structure

```r
ggplot()
```

Code structure

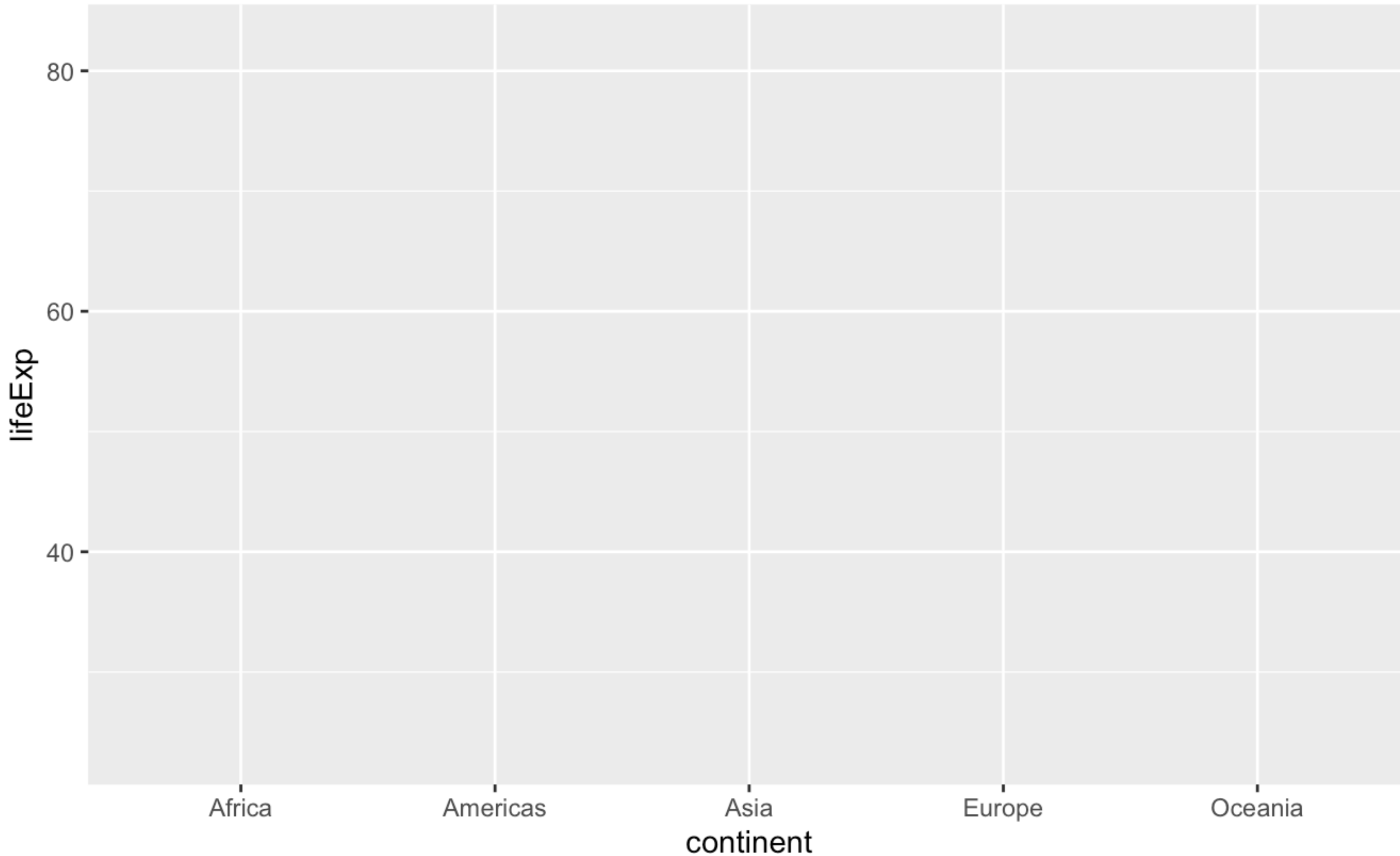
```
ggplot(data = gapminder)
```

Code structure

```r
ggplot(data = gapminder,
       mapping = aes())
```

Code structure

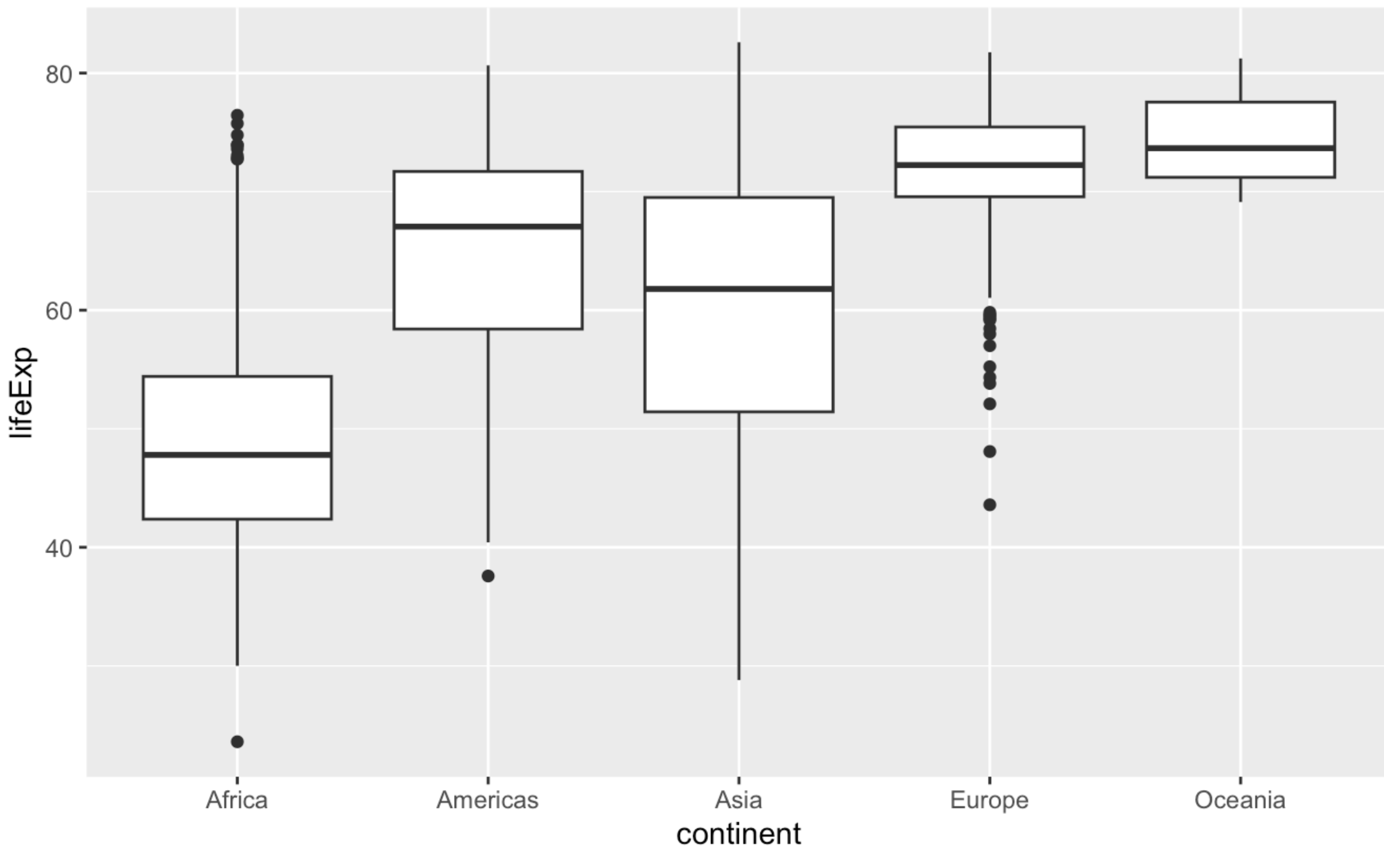
```
ggplot(data = gapminder,
       mapping = aes(x = continent,
                     y = lifeExp))
```



open
wash
data.

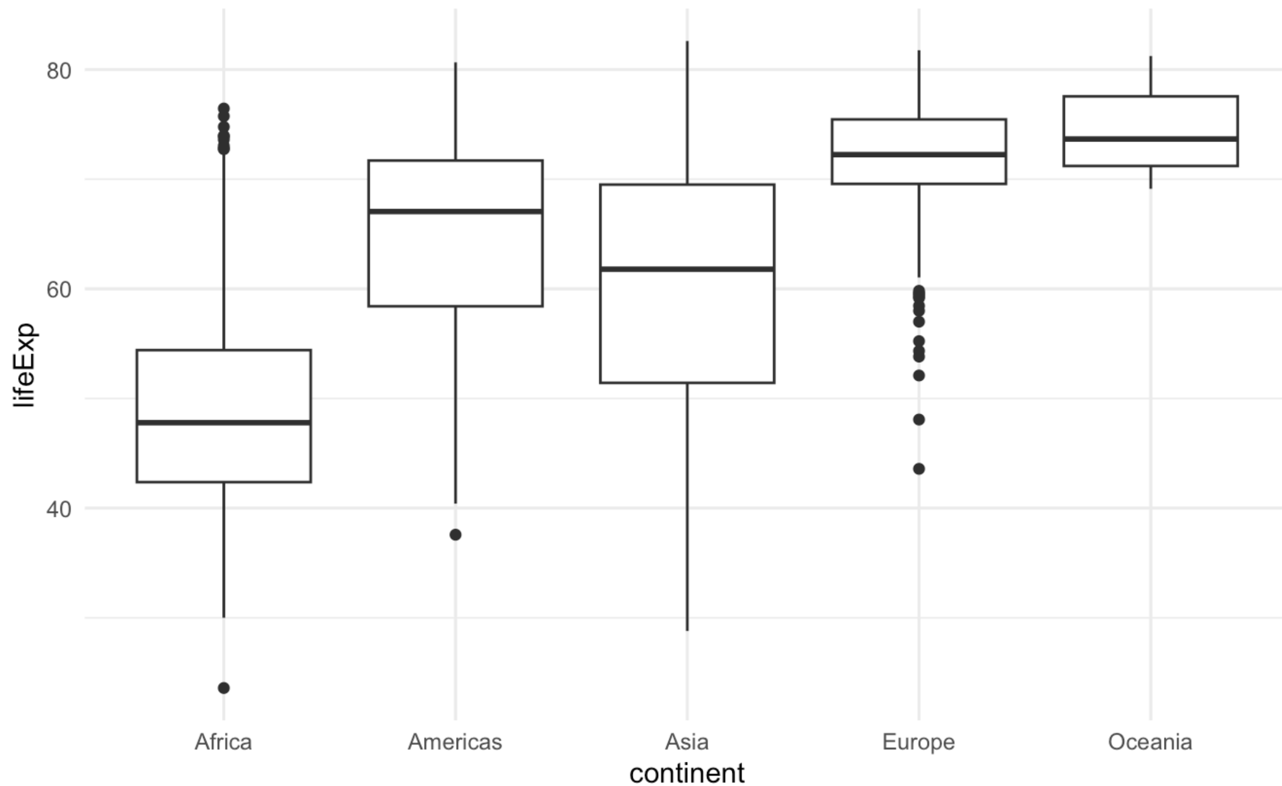
Code structure

```
ggplot(data = gapminder,
       mapping = aes(x = continent,
                     y = lifeExp)) +
  geom_boxplot()
```



Code structure

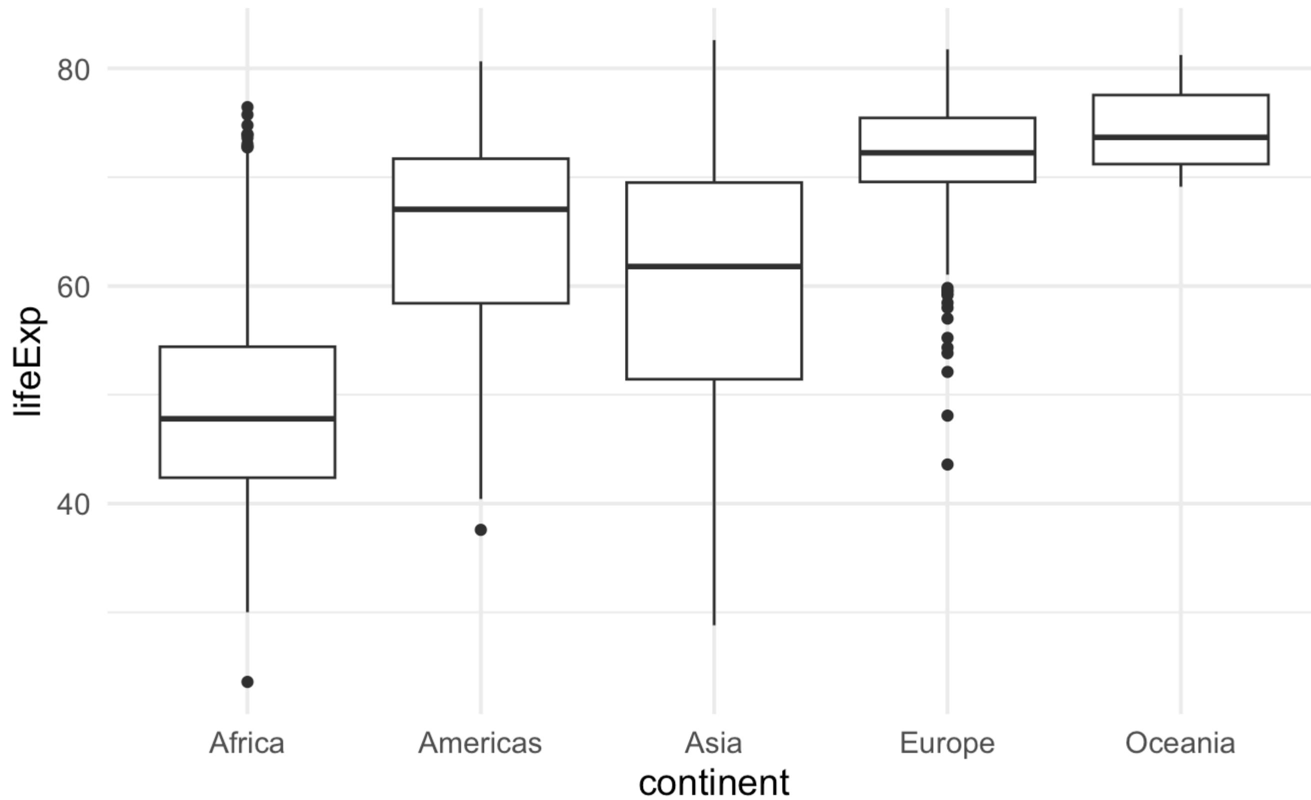
```
ggplot(data = gapminder,
       mapping = aes(x = continent,
                     y = lifeExp)) +
  geom_boxplot() +
  theme_minimal()
```



Polls

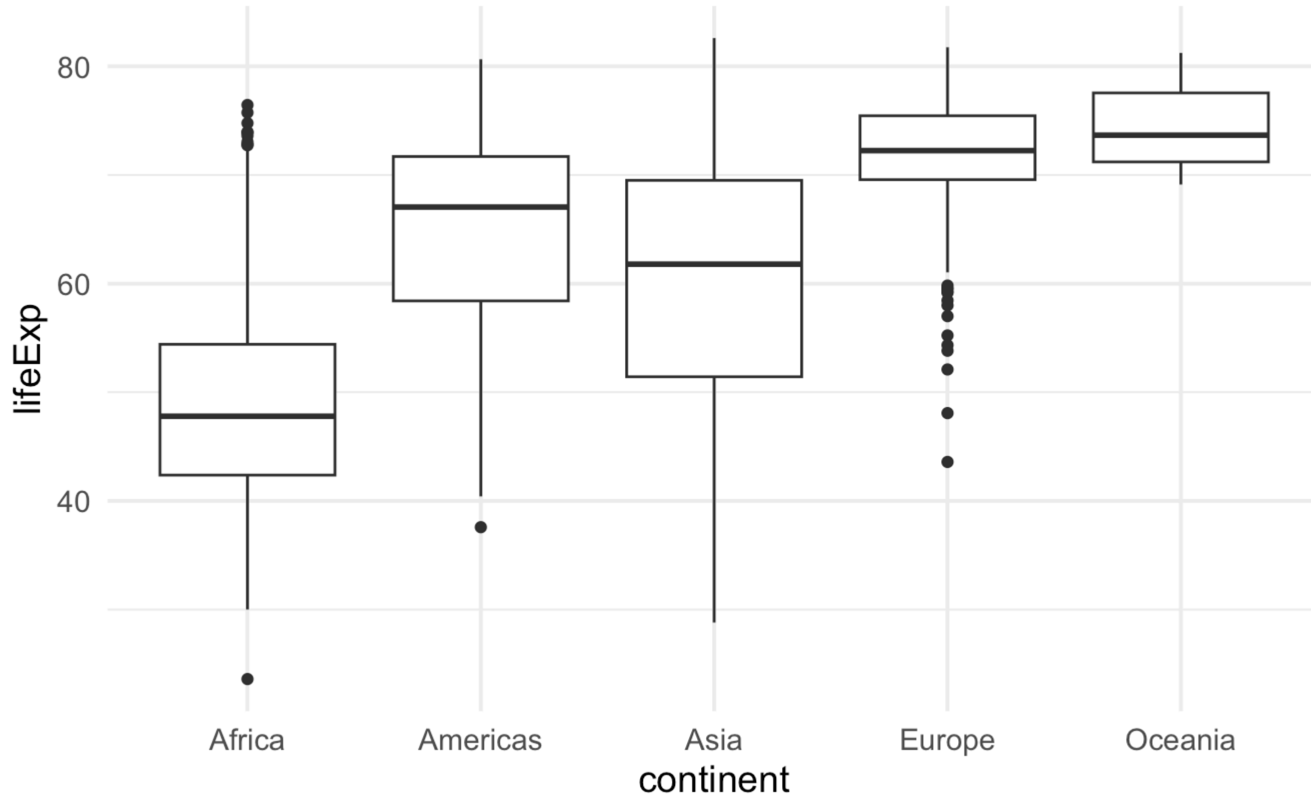
Poll 1: What does the thick line inside the box of a boxplot represent?

1. the mean of the observations
2. the middle of the box
3. the median of the observations
4. none of the above



Poll 2: What percentage of observations are contained inside the box of a boxplot (interquartile range)?

1. 25%
2. depends on the median
3. 50%
4. none of the above



Poll 3: What is the median of a set of observations?

1. The median is the most frequently occurring value in a dataset.
2. The median is the sum of all values in a dataset divided by the number of observations.
3. The median is the point above and below which half (50%) of the observations falls.
4. The median is the square root of the sum of the squares of each value in a dataset.

Boxplot, explained

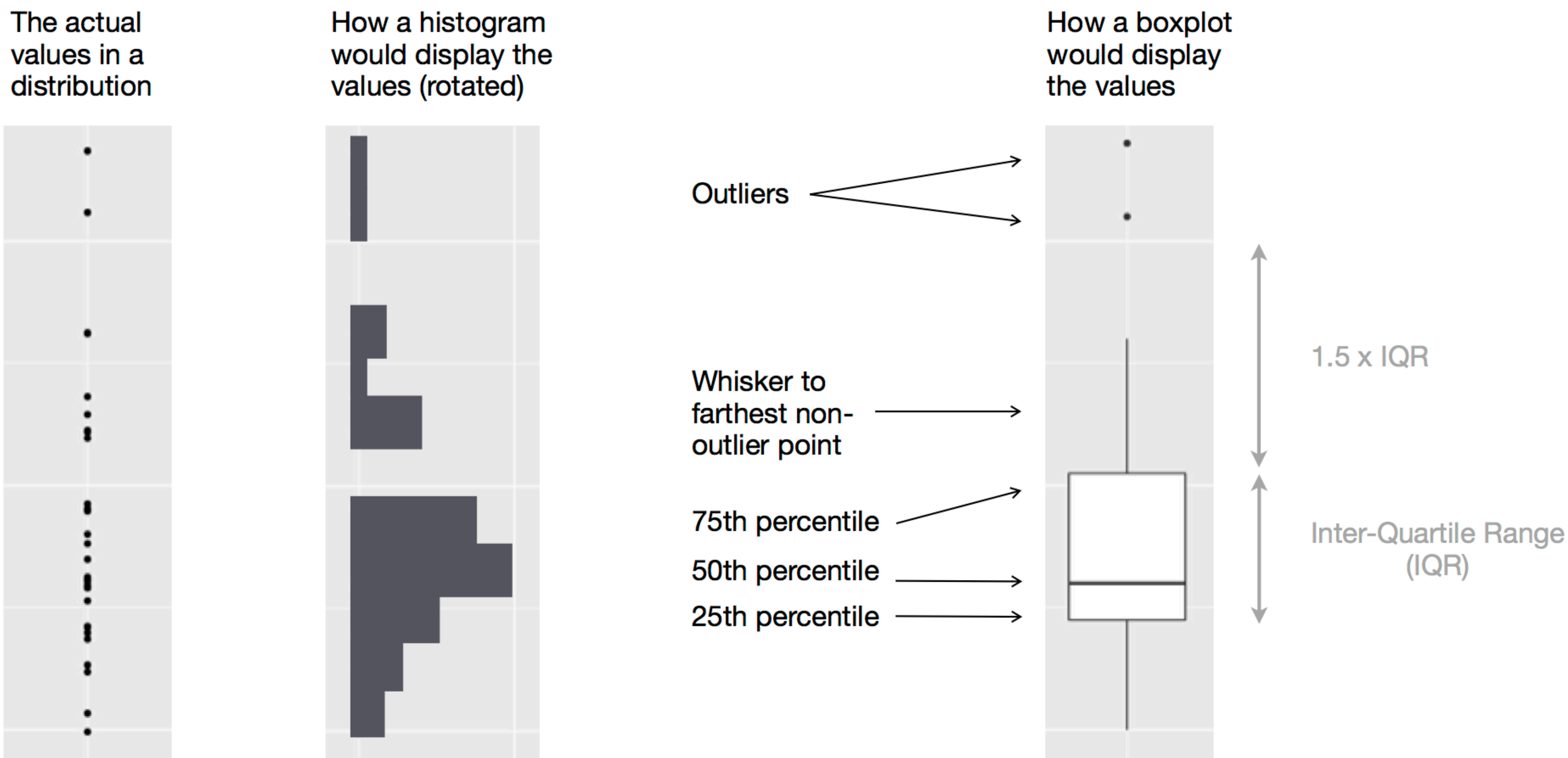


Figure 1: Diagram depicting how a boxplot is created.

Our turn: md-02-exercises

1. Open posit.cloud/spaces/663318/content in your browser.
2. Verify that the ds4owd workspace is open.
3. Click Start next to md-02-exercises.
4. In the File Manager in the bottom right window, locate the `md-02b-visualize.qmd` file and click on it to open it in the top left window.

30:00

Take a break

Please get up and move!

10:00

Visualizing data

Types of variables

numerical

discrete variables

- non-negative
- whole numbers
- e.g. number of students, roll of a dice

continuous variables

- infinite number of values
- also dates and times
- e.g. length, weight, size

non-numerical

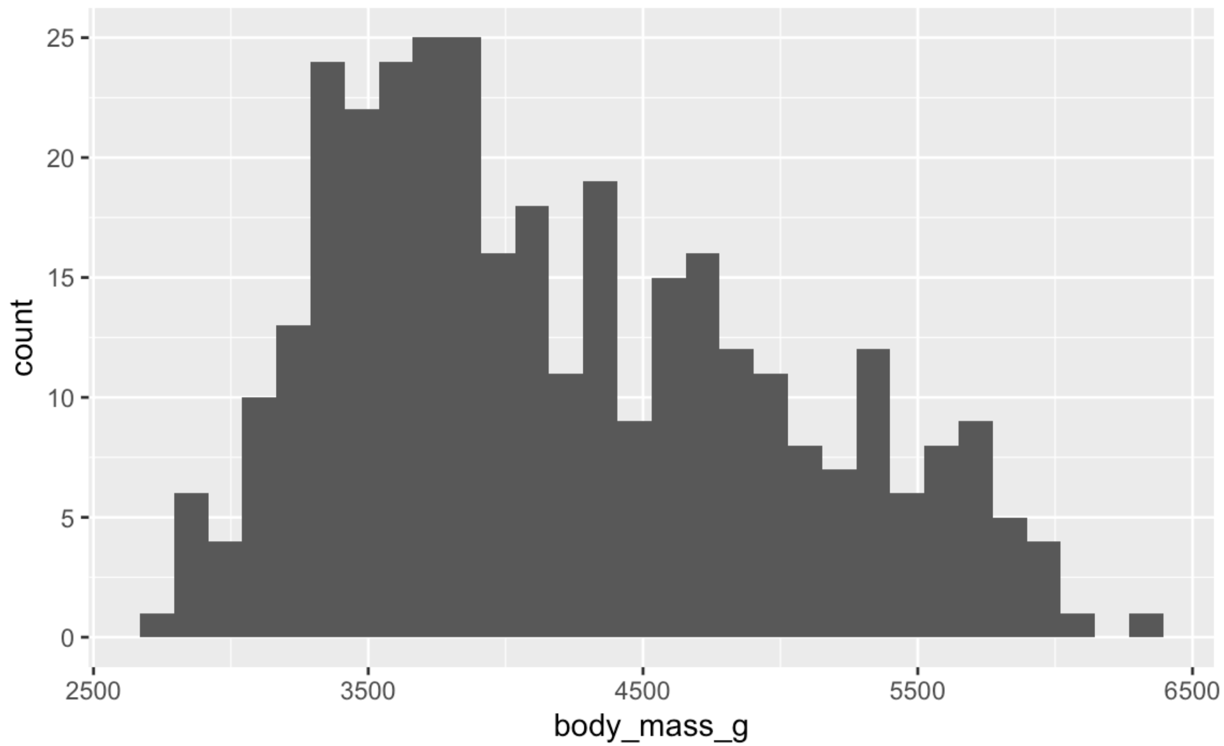
categorical variables

- finite number of values
- distinct groups (e.g. EU countries, continents)
- **ordinal** if levels have natural ordering (e.g. week days, school grades)

Histogram

- for visualizing distribution of continuous (numerical) variables

```r
ggplot(data = penguins,
       mapping = aes(x = body_mass_g)) +
  geom_histogram()
```



Barplot

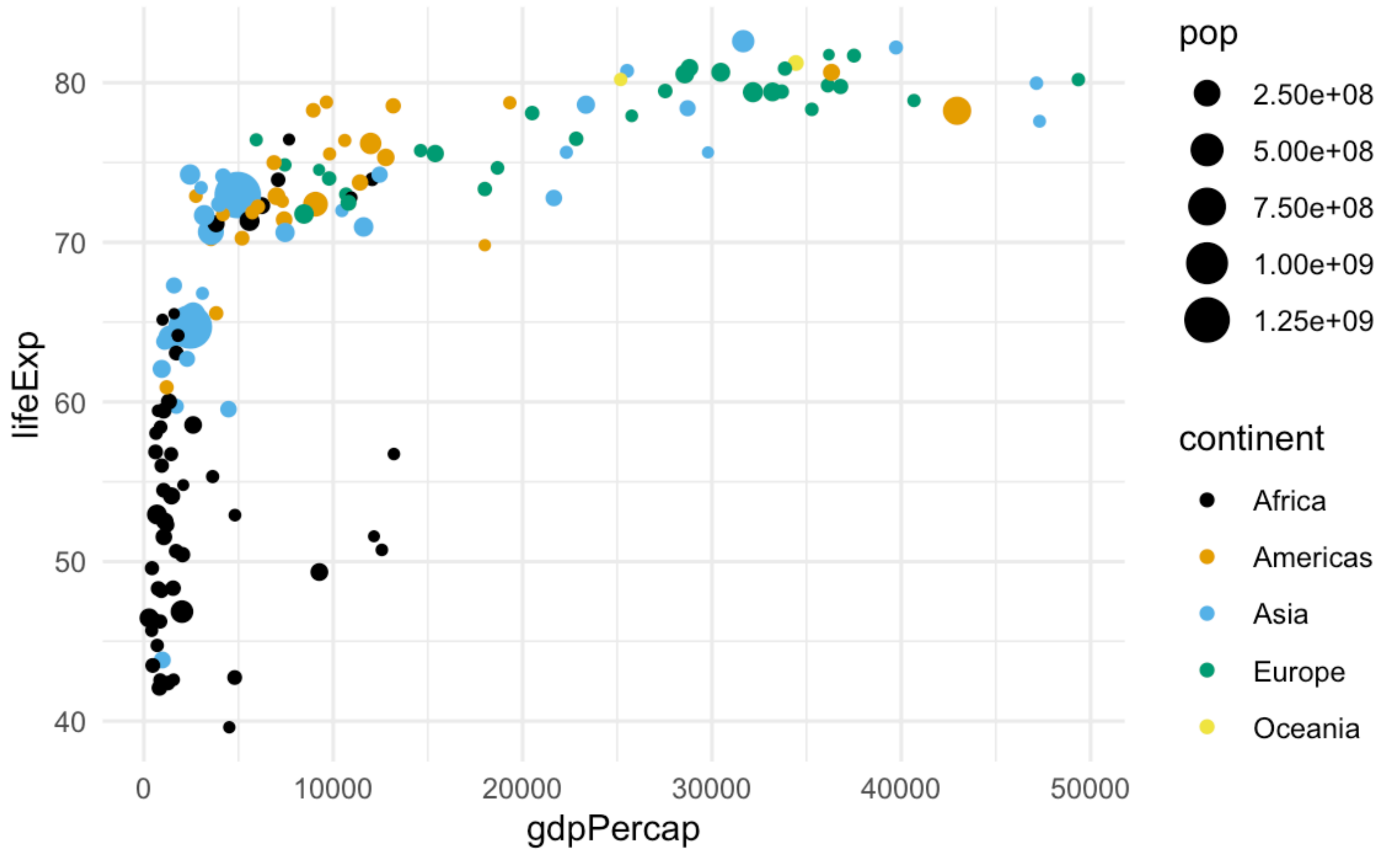
- for visualizing distribution of categorical (non-numerical) variables

```
ggplot(data = penguins,
       mapping = aes(x = species)) +
  geom_bar()
```

Scatterplot

- for visualizing relationships between two continuous (numerical) variables

```
ggplot(data = gapminder_2007,
       mapping = aes(x = gdpPercap,
                     y = lifeExp,
                     size = pop,
                     color = continent)) +
  geom_point() +
  scale_color_colorblind() +
  theme_minimal()
```



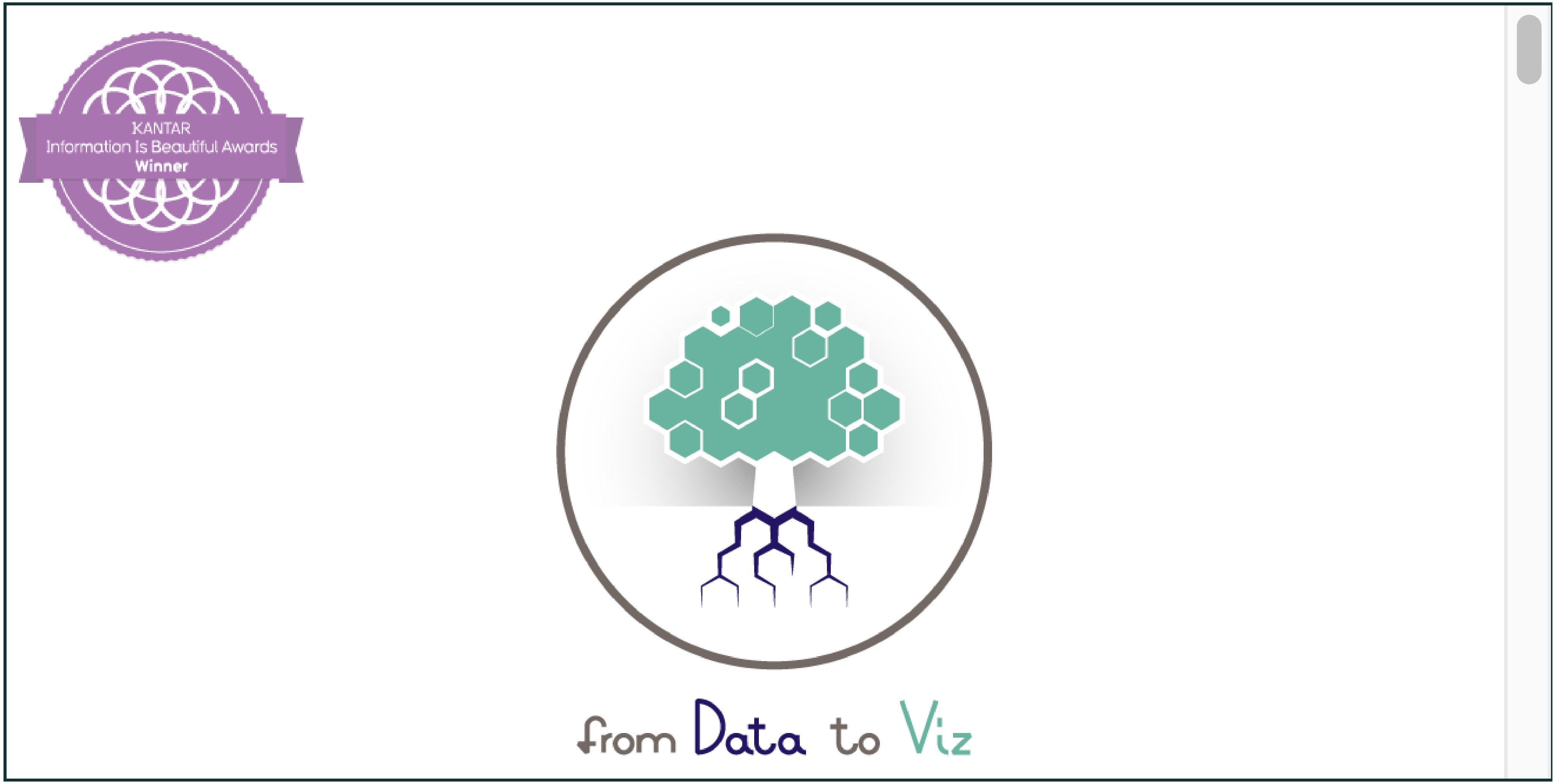
Your turn: md-02-exercises

1. If you closed it, open posit.cloud/spaces/663318/content in your browser.
2. Verify that the ds4owd workspace is open.
3. Click Continue next to md-02-exercises.
4. In the File Manager in the bottom right window, locate the `md-02c-boxplot.qmd` file and click on it to open it in the top left window.
5. Follow instructions in the file.

15:00

How to pick a plot?

data-to-viz.com



Homework assignments module 2

Module 2 documentation

ds4owd-002.github.io/website//modules/md-02.html

Module 2

Data science lifecycle & Exploratory data analysis using visualization

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Are you ready for some data visualisations? This week is all about exploring data with `ggplot2` R package. We will also learn about the data science lifecycle.

Learning Objectives

1. Learners can list the six elements of the data science lifecycle.
2. Learners can describe the four main aesthetic mappings that can be used to visualise data using the ggplot2 R Package.
3. Learners can control the colour scaling applied to a plot using colour as an aesthetic mapping.
4. Learners can compare three different geoms (bar/col, histogram,

Homework due date

- Homework assignment due: Wednesday, 2025-09-24
- Quiz due: Wednesday, 2025-10-01

Wrap-up

Thanks! 🌻

Slides created via revealjs and Quarto: https://quarto.org/docs/presentations/revealjs/

Access slides as PDF on GitHub

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